



2026: Steady, but AI & the Hawks are Circling (Software)

June 3, 2026

Over the last year, we have received a healthy dose of macro tailwinds. The rapid buildout of AI infrastructure, Fed & ECB rate cuts (with inflation still north of 2%), the announcement and enactment of German fiscal stimulus, and record high equity prices and near-record tight credit spreads have helped to boost risk sentiment.

Yet despite this, the credit default picture is far more mixed than appreciated. Yes, US spec-grade default rates did decline from a peak of 5.1% last July to 4% today, as cyclical risks remained muted, but current US default rates are well above 20year median levels of 2.9%, with CCC defaults rising over the past year. In Europe, the lack of AI tailwinds meant defaults actually increased from 2.1% to 3.8% in March & 4.6% in April, easily above 20 year median levels, with elevated defaults in cyclical sectors and in the challenged economies of France/UK/Germany offsetting strength in Southern Europe.

And this is the main message of our piece: It is true we aren't expecting a major global default cycle to emerge. But equally, don't let the aggregate data and market prices make you think dispersion is going to disappear. **US & European default rates should remain above long-term averages through Q2'27**, with low recovery rates for those names which default, and a continued presence of distressed exchanges highlighting that many capital structures remain overleveraged, especially now as central banks shift more hawkishly.

In the US, we expect spec-grade default rates to slow from 4% today to 3.6% by YE'26, and then to climb slowly again to 4% by Q2'27. In essence, this is a message that the K-shaped US economy still lives. As we have written repeatedly since 2024, the US economy is effectively a "fire & ice" US economy. Perhaps today, we can change that label to an "AI & ice" US economy, given how narrow we believe the drivers of US growth to be. **On the positive side, US artificial intelligence tailwinds** feed back into very strong US corporate profits and a wealth effect that supports the high-income US consumer, and this is a major reason why our US credit-based recession model is only forecasting a 12% chance of a US downturn through Q1'27. It is very hard to see a spike in US defaults with overall US growth so solid.

But equally, a look at the "typical" US business today shows margin compression and demand expectations at near the worst rate going back to 2022. A more micro look at \$HY coupons vs. \$HY revenue growth rates & \$HY debt to EV ratios tell a similar story. Default rates may not be alarming, but they are likely to stay more elevated than today's current spreads would imply.

And the tailwinds of AI adoption for aggregate growth are likely to be headwinds for spec-grade credit defaults through tighter financial conditions. This will eventually play itself out through two channels. The first will be more hawkish Fed policy and the second will be meaningful downgrades of US software names into CCC buckets, where investors have already

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demonstrated their impatience to provide flexibility to this rating cohort over the last few years.

The timing of this risk is admittedly uncertain, but it is growing more likely that a scenario could emerge in early 2027 where tangible AI disruption of software business models begins to damage software revenue and customer retention rates, at the same time Fed hikes emerge to eat up limited software free cash flow. This risk is getting priced in leveraged software names today, with 30% of US Software Lev Loans already trading with an implied CCC rating (vs. only 10% from the rating agencies). But a raft of software downgrades will have broader spillovers than appreciated. A quick spate of software downgrades is enough to push CCC buckets in US lev loan indices to 7.5%+, even assuming no broader sector risk. And further software stress is likely to increase liquidity pressure on BDCs, that could force the sale of leveraged loans and eventually force sizeable private credit loan markdowns. **With 80% of all \$B rated debt financed itself in either the Leveraged Loan or Private Credit market, this is clearly our biggest risk to monitor.**

In Europe, we forecast spec-grade defaults to decline from April's very high 4.6%, but to settle at 3.6% by YE'26 & 3.9% by Q2'27. Europe doesn't have a K-shaped economy, but a generally weak one that is getting worse. Our European credit-based recession model is now somewhat above average, as banks tighten lending standards on borrowers at the same time profit growth is sluggish. But this isn't the primary reason we are turning more cautious on European defaults relative to last year. The biggest change is that the ECB is now compounding the weak Eurozone growth with at least two rate hikes likely in 2026.

This flow-through of tighter ECB policy into European spec-grade credit should make an impact. B rated European bond and loan maturity walls are well above historical averages over the next 2-3 years, with 37% of B-rated European HY bond issuers today already free cash flow negative. This would rise to 44% if total funding costs rise 50bps; a plausible scenario in a world where the ECB is now set to hike 50bps, while coupons on maturing bonds 2-3 years out, are 70 to 170bps below current market yields. Leveraged European businesses need lower rates before upcoming maturities, but it is now hard to see this materializing.

In the rest of the piece, we discuss which US & European sectors are most at risk of defaults, while assessing which rating buckets are appropriately or inappropriately priced for the risks ahead.



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Summary Bullets

Default Study: Steady, but AI & the Hawks are Circling (Software)

- **The main message of our piece:** It is true we aren't expecting a major global default cycle to emerge. But equally, don't let the aggregate data and market prices make you think dispersion is going to disappear. **US & European default rates should remain above long-term averages through Q2'27**, with continued pressure on global CCCs, and low recovery rates for those names which default.
- **In the US**, we expect spec-grade default rates to slow from 4% today to 3.6% by YE'26, and then to climb slowly again to 4% by Q2'27. In essence, this is a message that the K-shaped US economy still lives. Robust AI tailwinds will prevent a major spike in defaults, but businesses flagging margin compression & demand weakness ahead plus the prospect for further increases in \$HY bond yields will prevent defaults from falling much as well.
- **Software downgrades are the largest risk for US spec-grade credit today.** If AI disruption shows up in weaker software fundamentals (rather than just lower multiples), software downgrades will fill up CCC buckets in US Lev Loan indices and force through meaningful private credit loan markdowns. With 80% of all B-rated debt in US spec-grade financed via private credit or leveraged loans, this risk is material, but timing of it is far from trivial.
- **In Europe**, we forecast spec-grade defaults to decline from April's very high 4.6%, but to settle at 3.6% by YE'26 & 3.9% by Q2'27. Europe doesn't have a K-shaped economy, but a generally weak one that is getting worse, with cyclical defaults rising, and upward pressure placed on defaults from sluggish growth in France, the UK & Germany. But a shift from expecting ECB cuts to ECB hikes is the key reason we are shifting more cautiously on Europe. 44% of European B rated HY bond issuers will be in negative free cash flow if total funding costs rise 50bps; a plausible scenario in a world where the ECB is set to hike 50bps, while coupons on maturing bonds 2-3 years out, are 70 to 170bps below current market yields.
- **Which US sectors could be most at risk for defaults?** The \$HY sectors with the highest leverage currently are Transports, Media, and Telecoms. Telecoms are well flagged, with a high share of current distressed debt. Consumer Goods and Healthcare leverage is high in their respective histories and are likely to come under further pressure even in a mild shock. If the ongoing commodity shock impacts EBITDA margins more substantially and rising rates cause multiples to re-rate lower, Cyclical sectors such as Leisure, Transports, Retail, Tech, Capital Goods will come under greater pressure.

What's Priced into Credit?

- **Distressed debt risk is most concentrated in USD Spec-Grade markets**, particularly within Technology/Software and Telecom sectors. Over a third of distressed \$HY debt is in Telecoms, with Technology the second largest risk at just under a sixth of distressed bonds. \$Leveraged Loan distressed debt is dominated by Software (nearly two-fifths of distressed debt, three times its overall index weight), while **risk in €HY and €LL is more dispersed across both Cyclical and Non-Cyclical sectors**. Wireless, Healthcare, and Non-Electric Utilities lead among Non-Cyclicals, along with significant debt levels in Cyclical sectors like Consumer Cyc Services, Chemicals, and Packaging. In €LL, Trading Companies/Distributors and Healthcare each hold a 20% share of distressed loans, with Real Estate and Software also



notable contributors.

- **\$HY and \$LL distress ratios started to diverge in October of last year** as investors became more wary of private credit risks. The level of distressed \$LL issuers rose as AI-disruption became a theme early this year. Through all of this, \$HY distress ratios have remained relatively resilient with more imminent risks priced into the loans market. €Spec-Grade distress ratios are elevated compared to post-European Debt Crisis lows, reflecting an expectation for moderate defaults moving forward.
- IG rating bands, as expected, continue to compensate buy-and-hold investors for all observed historic default periods. **Current BB & B spreads would compensate investors** for default risk in most time periods historically outside of protracted recessions. \$CCCs are unattractive under any scenario other than among the best (lowest) outlooks for defaults seen in history with strong recovery rates.
- Assuming a 40% recovery rate, **current HY index spreads would not have compensated investors in 11 of the previous 40 cohorts**, which are all grouped around '85-'90 and the dot-com bubble burst. Even at a low recovery rate, BB investors would be compensated in nearly all cohorts, even if recovery rates were lower the average. For a buy-and-hold investor, current CCC spreads only compensate bond holders in very low default environments with strong recovery rates.

Data as of COB 05/28/2026

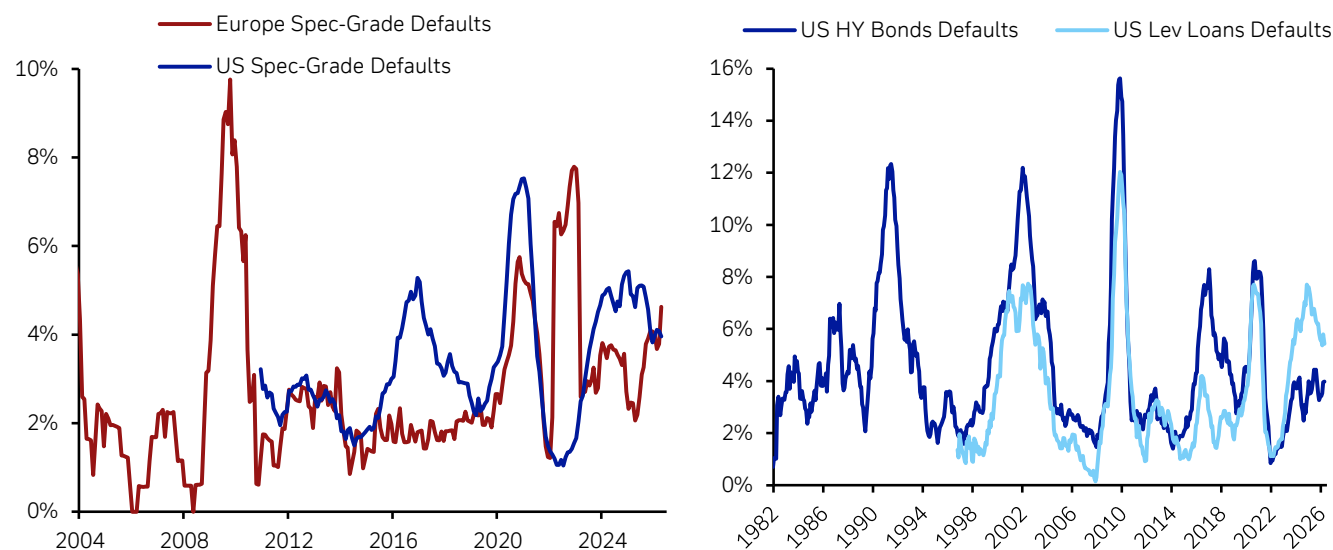


US spec-grade default rates

Better than expected, but no return to a low default world

US spec-grade default rates have made a steady, albeit unspectacular, move lower since we published our 2025 default study. Trailing 12m issuer-weighted US speculative-grade default rates (including distressed exchanges) are 4%, near the lows seen since 2023, and down from a peak rate of 5.1% reached last July. Nearly all of this improvement has come from the \$-Lev Loan market, where default rates have dropped 120bps from last July. By contrast, \$HY bond default rates have been far steadier, only dropping around 20bps from last July, and climbing again in recent months ([Figure 1](#)).

Figure 1: \$-Speculative-grade, €-Speculative-grade, \$HY and \$-LL default rates (all include distressed exchanges, trailing 12m)



Source : Deutsche Bank, Moody's, S&P

The drop in US spec-grade default rates belies levels that are still fairly elevated in historical context. Case in point: The median & average US spec-grade default rate over the last 20 years is 2.9% & 3.6% respectively, vs. 4% today. Hence, US credit stress is actually higher than what many investors have grown accustomed to, especially in CCC-rated bonds & loans, where little leeway is being given for over-leveraged capital structures. But equally, default rates have fallen further than we predicted in our 2025 study, with our projection of 4.8% by Q2'26 proving too pessimistic.

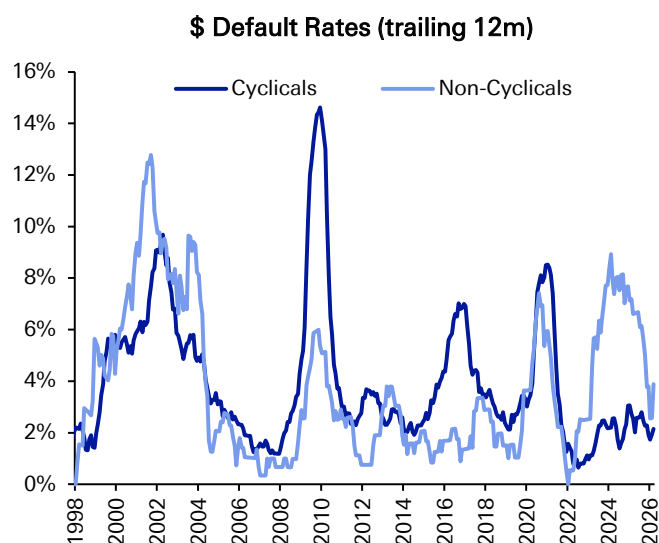
Over the last year, US defaults have been nudged lower by a stronger US economy, led by continued AI tailwinds, but also an accommodating Federal Reserve, and certainly a Fed that was more accommodating than we had expected. In our view, a world of above-target inflation & severe policy uncertainty (due to US tariffs) meant the Fed would not provide rate cuts, before job cuts. However, the reality was different. The Fed cut rates 75bps despite US core PCE running at a steady 2.8% Y/Y



pace and with little tangible realisation of meaningful job layoffs.

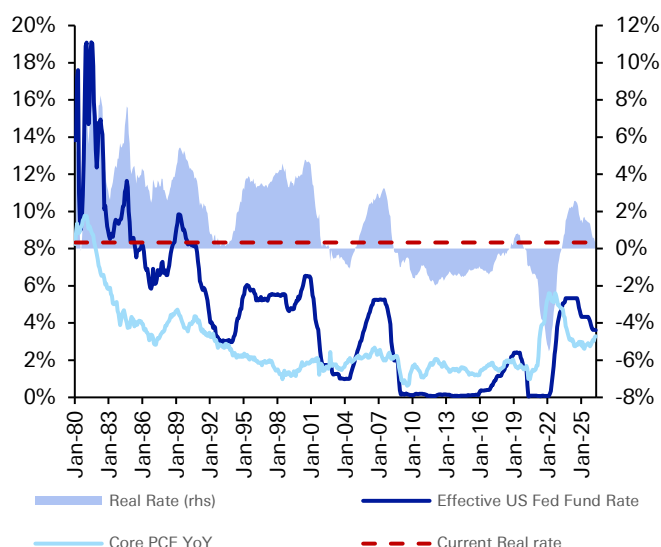
The result was certainly a better mix of cyclical & non-cyclical default rates than we anticipated. Fed insurance cuts and lower overall yields, plus past restructurings in the Telecom & Healthcare sector, allowed non-cyclical default rates to drop materially over the last year, while cyclical defaults remained unchanged as the US unemployment rate remained surprisingly steady in the current “low-hire, low-fire” business environment ([Figure 2](#)).

Figure 2: USD Non-Cyclical defaults slowed over the last year while Cyclical defaults were steady



Source : Deutsche Bank, S&P
*issuer-weighted

Figure 3: US real rates moved lower over the last year, as the Fed did cut before job cuts

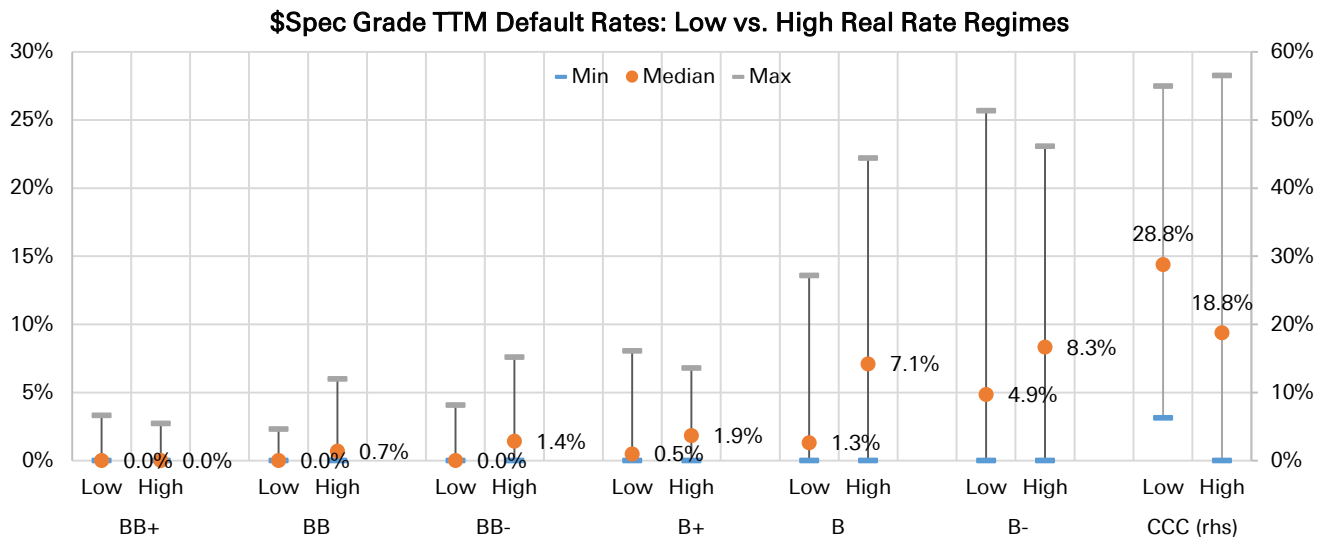


Source : Deutsche Bank, Bloomberg Finance LP

In addition, real US interest rates continued their decline seen over the last 2 years ([Figure 3](#)). And as we wrote in our 2024 default study, the trajectory of real interest rates is critical for the trajectory of future default rates, particularly for B2 & B3 rated bonds and loans. In [Figure 4](#) below, we produce a similar exercise, dividing up US economic history into low and high real rate regimes, as measured by months in which the difference between the effective Federal Funds Rate and Core Y/Y PCE is in its bottom third or top third percentile, since 1980. As [Figure 4](#) illustrates, the median B2 & B3-rated annual default rate is far more elevated in a high real rate (7.1% & 8.3%) vs. a low real-rate regime (1.3% & 4.9%). And last year’s data corroborates this trend. US B-rated default rates were well below recent averages, and certainly far closer to a low real-rate than high real-rate world (Fig 5). Simply put, Fed policy was a tailwind for Bs last year.



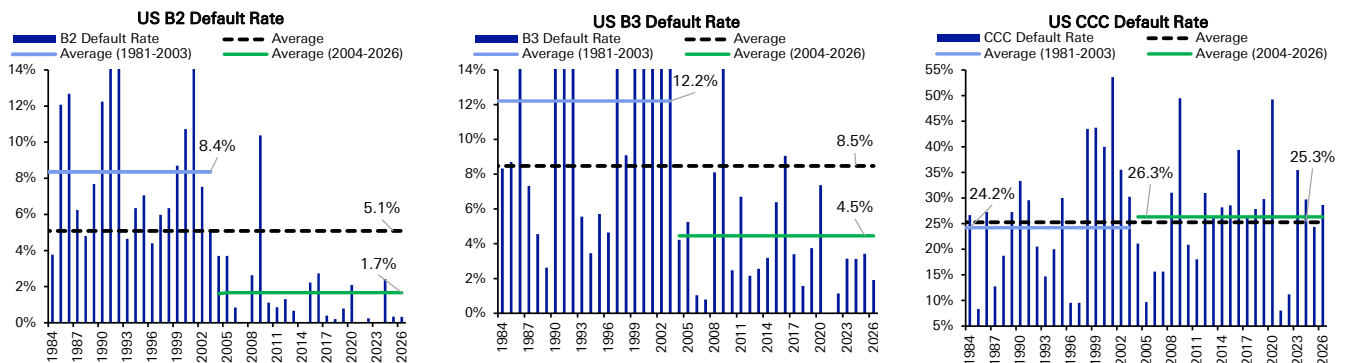
Figure 4: The Fed's next move matters. Real rates make a big difference in B2 & B3 default rate trends.



Source : Deutsche Bank, Bloomberg Finance LP, S&P, \$-spec grade default rates are available from Dec'81 onwards. We define the High (Low) real rate regime as the months in which the effective Federal Funds rate - core Y/Y PCE is in the top (bottom) 1/3 of its distribution since 1980.

Note however that real rates have a negligible impact on CCC default rates, both historically, and also last year, when US CCC default rates were quite elevated, and in line with historical averages. This is certainly a cautionary tale for future US default trends. Easier Fed policy and/or macro tailwinds from a transformative AI buildout will not prevent overleveraged capital structures from restructuring in secularly challenged industries. We have already experienced this directly in sectors such as US Telecom & US Healthcare in recent years. But one can apply this framework to US Software, especially considering the market is pricing 30% of US Software Lev Loans today with an implied CCC rating (more on this later).

Figure 5: The Fed cutting with inflation at 2.8% and limited job cuts kept B default rates at very low levels. But CCC default rates remain elevated, a cautionary tale for leveraged issuers that run into financial trouble ahead.



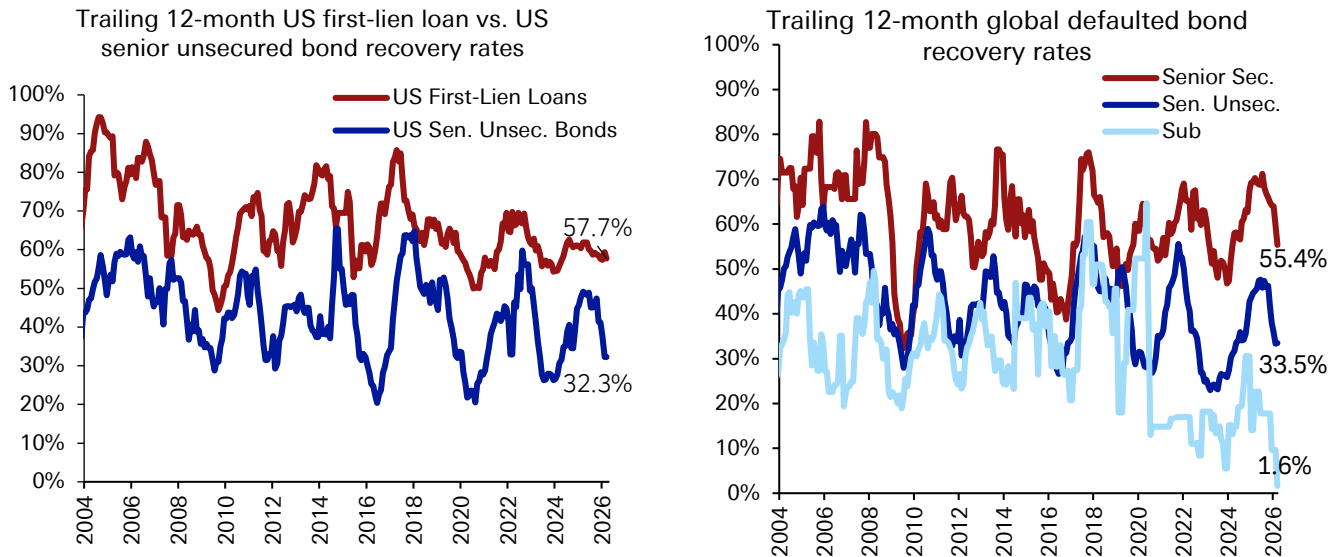
Source : Deutsche Bank, S&P

Despite rate cuts and a drop in US defaults, our views on lower recoveries & the continued prevalence of distressed exchanged did play out. One can see in Figure 6 that trailing 12m US Lev Loan recovery rates are only around 58%, while US HY senior unsecured bond recovery rates are stuck at a very low 32%. The picture is similar on a global basis, and when considering subordinated spec-grade debt as



well, where recovery rates are near 2% over the last year. Overleveraged capital structures, with often little subordinated debt cushions, and often high levels of intangible assets as collateral are unsurprisingly seeing poor recovery rates.

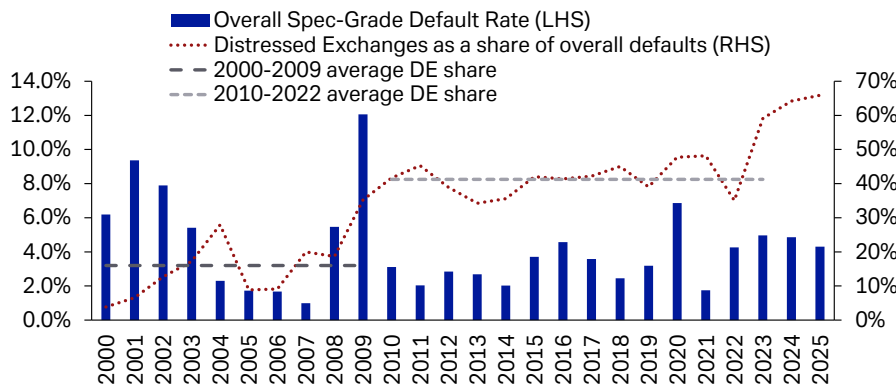
Figure 6: Default rates have been less severe than we expected. But recovery rates have been trending lower.



Source : Deutsche Bank, Moody's
*Measured by debt prices which are taken immediately prior to distressed exchanges or 30 days after non-distressed exchange defaults

On the distressed exchange point, while we have seen some slowdown in the monthly series we highlight in our Global Monthly Default Monitor, the overall prevalence of distressed exchanges remains. Distressed exchanges are an alternative to outright defaults or bankruptcies, in which creditors agree to restructure to continue receiving interest payments at the highest recovery rate possible, while the owners generally continue to retain their equity upside in the firm. In looking at data from Moody's, distressed exchanges made up 68% of overall global spec-grade defaults last year after being just under 60% of defaults in 2023. The post-GFC average until 2022 is just over 40% of defaults, while prior to 2009 LME activity was relatively rare with an annual average of around 16% (Figure 7).

Figure 7: Distressed exchanges made up the vast majority of 2025 USD defaults



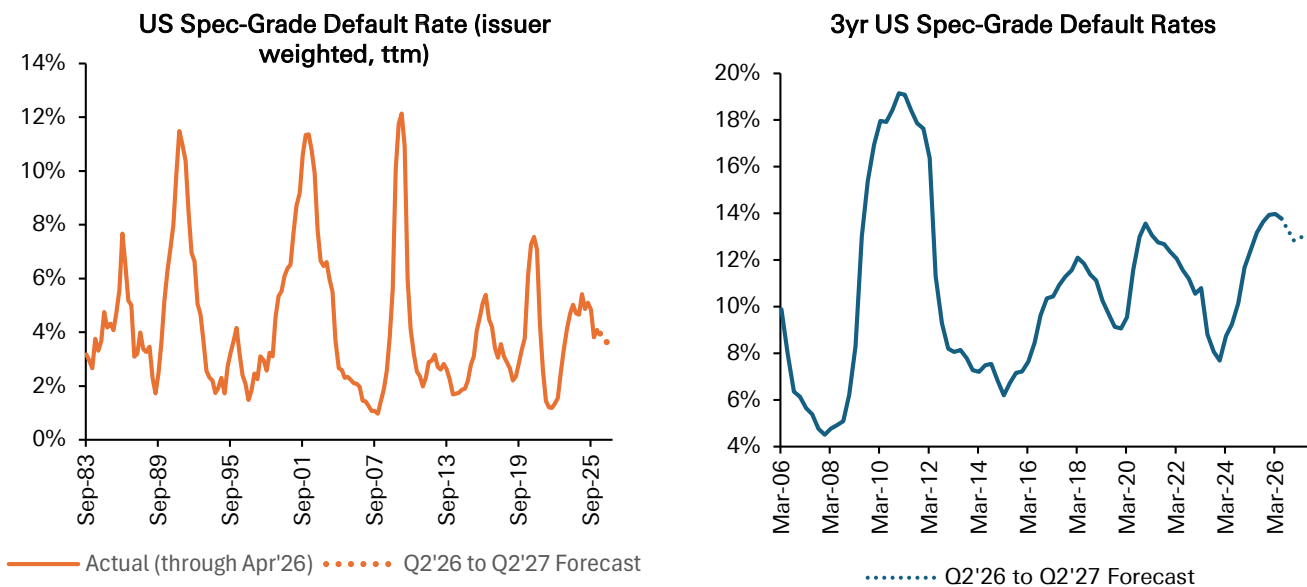
Source : Deutsche Bank, Moody's



What is the future outlook for US defaults?

We aren't forecasting a major US default cycle to emerge by the middle of 2027. But equally, the grind lower in defaults seen over the last year should abate; a return to say 2% defaults that were seen in past exuberant market cycles such as 2007, 2014 & 2021 is highly unlikely. Our forecast is for US spec-grade default rates to slow from 4% today to 3.6% by YE'26, and then to climb slowly again to 4% by Q2'27. If we are correct, the cumulative 3yr US spec-grade default rate will still be near some of the highest levels seen since the aftermath of the GFC ([Figure 8](#)).

Figure 8: US spec-grade defaults should fall from 4.0% to 3.6% by YE'26, and then slowly climb to 4.0% by Q2'27. We don't see a broad default cycle. But we also don't see default rates collapsing, given higher rates and pockets of leverage. This is why 3yr cumulative default rates will still remain elevated.

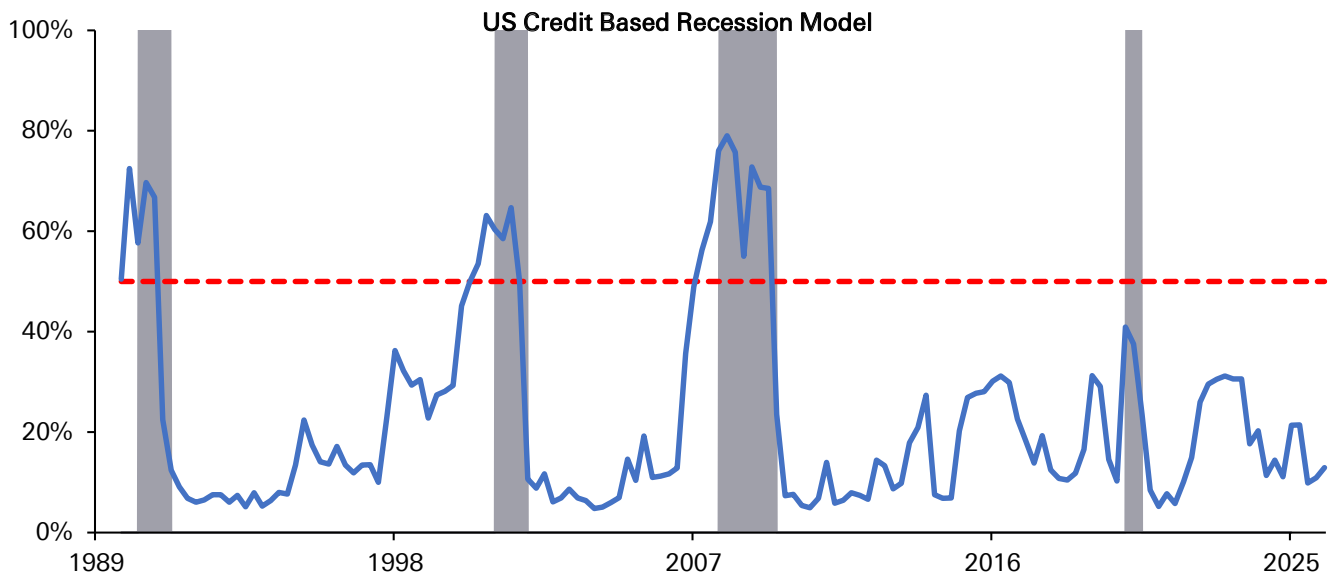


Source : Deutsche Bank, Bloomberg Finance LP, S&P

In essence, this is a message that **the K-shaped US economy still lives**, and that US credit dispersion will persist. As we have written repeatedly since 2024, the US economy is effectively a "fire & ice" US economy. Perhaps today, we can change that label to an "AI & ice" US economy, given how narrow we believe the drivers of US growth to be. On the positive side, US artificial intelligence tailwinds feed back into very strong US corporate profits and a wealth effect that supports the high-income US consumer, and this is a major reason why our US credit-based recession model is still only forecasting a 12% chance of a US downturn through Q1'27. It is very hard to see a spike in US defaults with overall US growth so solid ([Figure 9](#)).



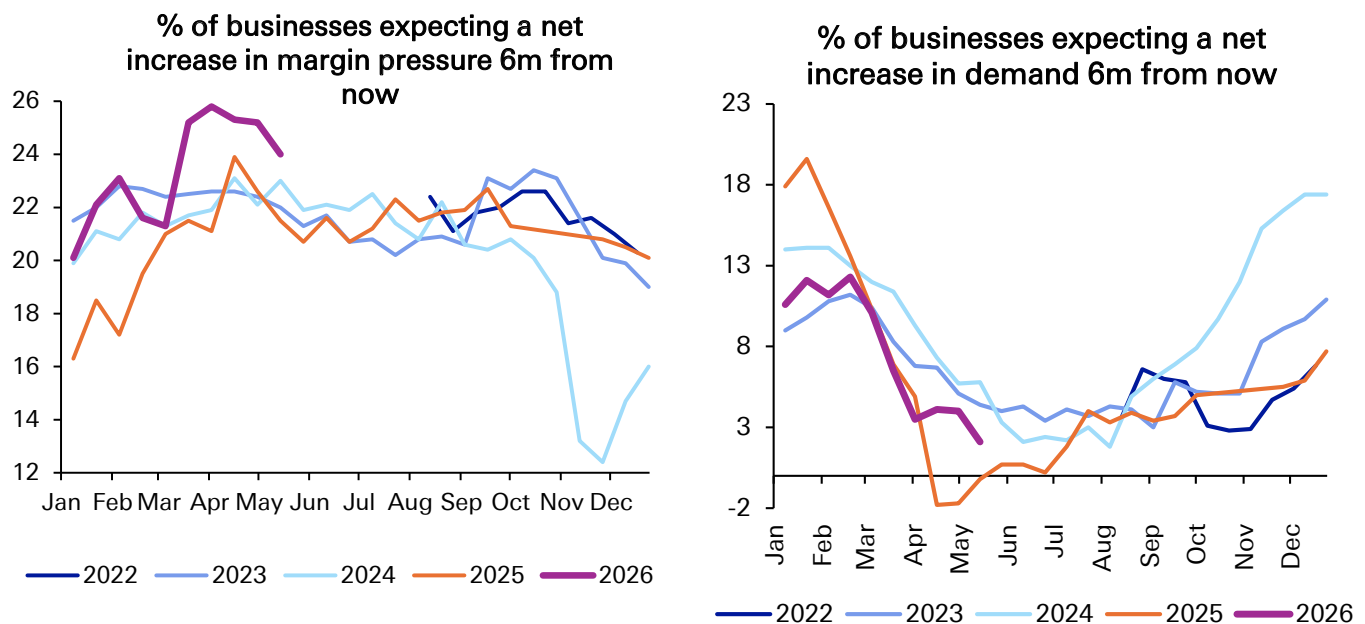
Figure 9: US cycle risks remain low, given robust corporate profit growth in particular.



Source : Deutsche Bank, Bloomberg Finance LP, Haver, "Our US & EU credit indicators take an average of 0-12m forward recession probabilities across the following 4 indicators: 1) Y-Y change in GDP-based corporate leverage, 2) Y-Y change in GDP-based corporate interest coverage, 3) Y-Y change in US bank loan 30-89 day past due rates or Eurozone bank NPL rates, 4) SLOS Survey: % of banks tightening standards on small firm C&I loans or ECB BLS survey: % of banks tightening standards across business/mortgage/consumer loans.

But equally, a look at the median US business, as proxied by the US Census Business Trends & Outlook Survey, paints a far different picture. US businesses today are forecasting future margin compression and future demand expectations at near the worst rate in 5 years (Figure 10). This would certainly imply higher \$HY & \$Lev Loan defaults, than what the aggregate macroeconomic data would imply.

Figure 10: Aggregate economic data looks strong, but a look at the "typical" US business shows considerable worry over future margin pressure and future demand trends. The K-shaped economy still lives.

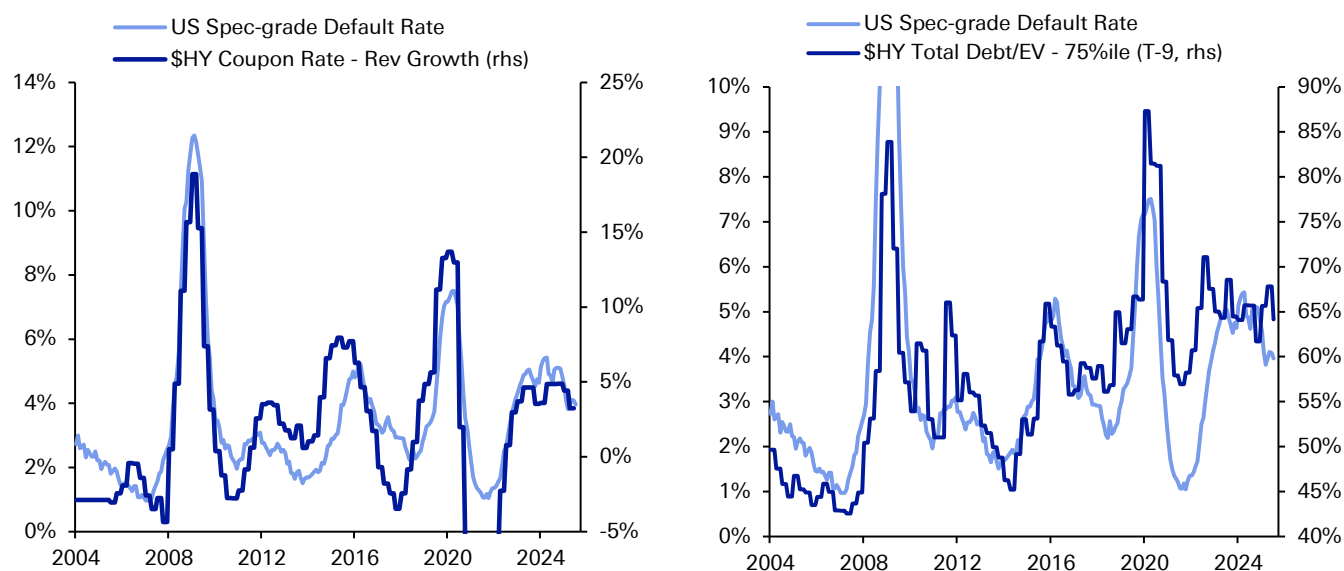


Source : Deutsche Bank, Business Trends & Outlook Survey



And a couple of micro overlays also suggest US spec-grade defaults will remain steady, but more elevated than what current spreads imply. The differential between \$HY index coupons & \$HY revenue growth rates has had a very strong relationship with contemporaneous defaults. Conceptually, if corporate debt service payments are rising faster than corporate income, it is harder for a company to service its debt. Currently, the \$HY index coupon is 6.6% vs. a median company revenue growth rate of 3.4%, suggesting default rates will remain elevated, but should settle around 4.0%. Similarly, \$HY corporate debt to EV ratios have strongly tracked future defaults, 9 months forward. This logic follows a Merton-Model style framework, where a sharp decline in equity valuations can signal the value of a firm's assets is at risk of declining below the value of a firm's debt, creating incentives for corporate owners/sponsors to restructure, to the detriment of creditors. In [Figure 11](#), we compare the 75%ile of \$HY debt to EV ratios vs. default rates, focusing on the 75%ile to capture the most leveraged public firms in the \$HY universe. This analysis is also consistent with US spec-grade default rates settling around 4.0% by year-end.

Figure 11: \$HY coupon - revenue growth & \$HY debt to EV ratios are largely consistent with our macro models forecasting 4% US spec-grade defaults over the next 9 months. Again, fundamentals do not suggest a major default cycle, but nor do they suggest a return to the low default rate world.



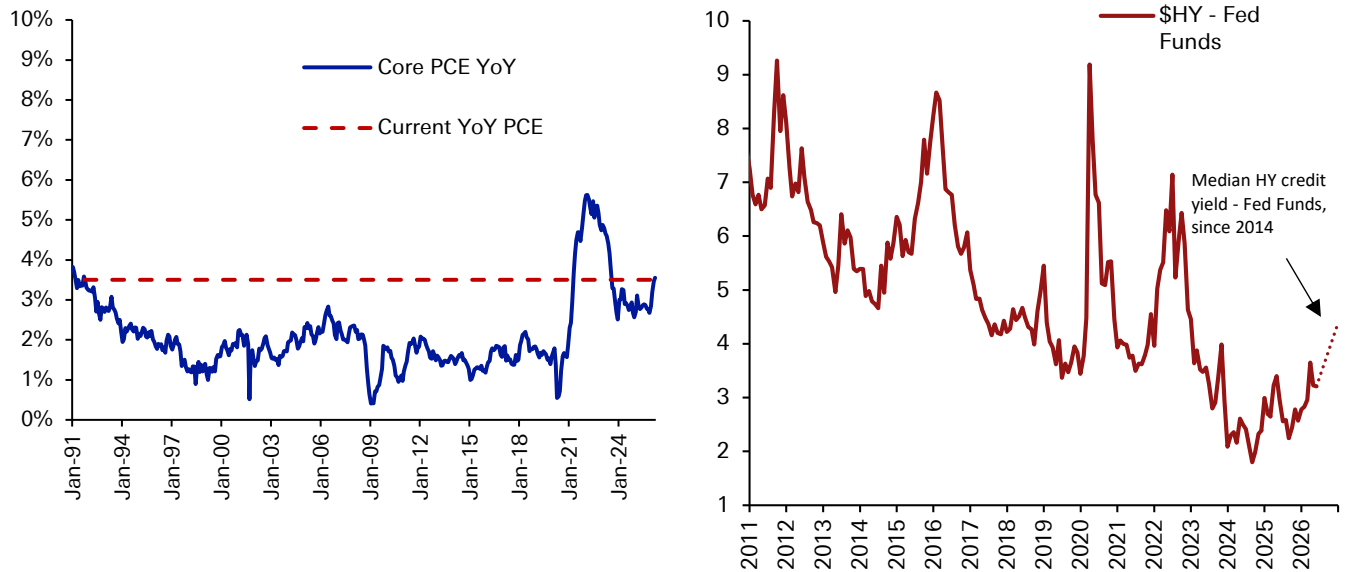
Source : Deutsche Bank, Bloomberg Finance LP, S&P

Lastly, **Fed policy is unlikely to be as accommodative** for US spec-grade issuers going forward, and the continued adoption of Artificial Intelligence is the reason why. US core PCE inflation is still running around a 3.5% Y/Y pace, which obviously reduces the odds of a Fed easing bias on its own. This is important considering overall \$HY bond yields still trade 120bps too low relative to the current Fed Funds rate on an historical basis ([Figure 12](#)). In the context of low recession risks + a long Fed hold, \$HY bond yields are likely to rise as US Treasury curves steepen. But even the risks of Fed hikes are starting to emerge, led by AI-associated capex and increasing memory chip shortages that are helping to keep US consumer inflation above target. The worst case scenario for US spec-grade credit markets and default rates would be Fed rate hikes starting at year-end 2026 or early 2027, at the same time that AI adoption is more tangibly disrupting the business model of floating-rate



software issuers in the Lev Loan & Private Credit markets. This latter scenario is not in our base case, as we expect the Fed to resist rate hikes for as long as possible, but it is certainly the largest risk in our view, and one we explore more fully later in this piece.

Figure 12: US core inflation is well above target, and \$HY almost looks like it is priced for Fed cuts (rather than hikes), with \$HY bond yields currently trading 120bps expensive to the Fed Funds rate, since 2014.



Source : Deutsche Bank, Bloomberg Finance LP



European spec-grade default rates

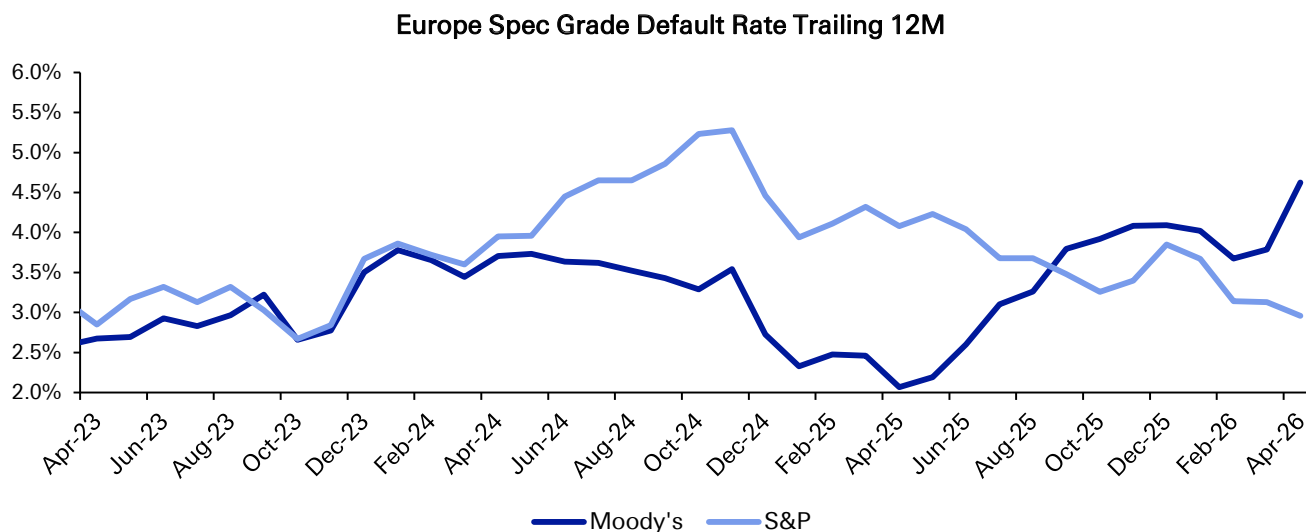
Surprise increase in defaults signals vulnerability to ECB hikes.

In contrast to the US, European spec-grade default rates have been climbing fairly aggressively over the last year. Moody's European spec-grade defaults have increased from 2.1% last year to 3.8% in March and 4.6% in April, after a particularly elevated month of April credit events that was the highest since Dec'23. European corporate default rates are now well above historical levels, with the median & average spec-grade default rate at only 2.3% & 2.9% over the past 20 years.

Our forecast last year for European default rates to stay stable, with levels around 2.3% by Q2'27, proved to be far too optimistic. What happened in Europe to push default rates so much higher?

The first item to note is that smaller sample sizes in Europe can make a large difference, when comparing default rates between different rating agencies. For example, S&P European spec-grade default rates have actually been declining gradually to around 3% in recent months ([Figure 13](#))

Figure 13: European spec-grade default rates are diverging between Moody's & S&P



Source : Deutsche Bank, S&P, Moody's

However, this is not necessarily a signal that European default rates will converge lower from here. On the contrary, one can make an argument that a higher Moody's default rate might be a leading signal for a higher future S&P default rate in the coming months. In our view, a higher Moody's European default rate is reflecting the fact that 1) Moody's takes a more conservative view than S&P with respect to what constitutes a credit default & 2) most defaults have been in the form of distressed exchanges, which Moody's is simply faster to recognize than S&P is.

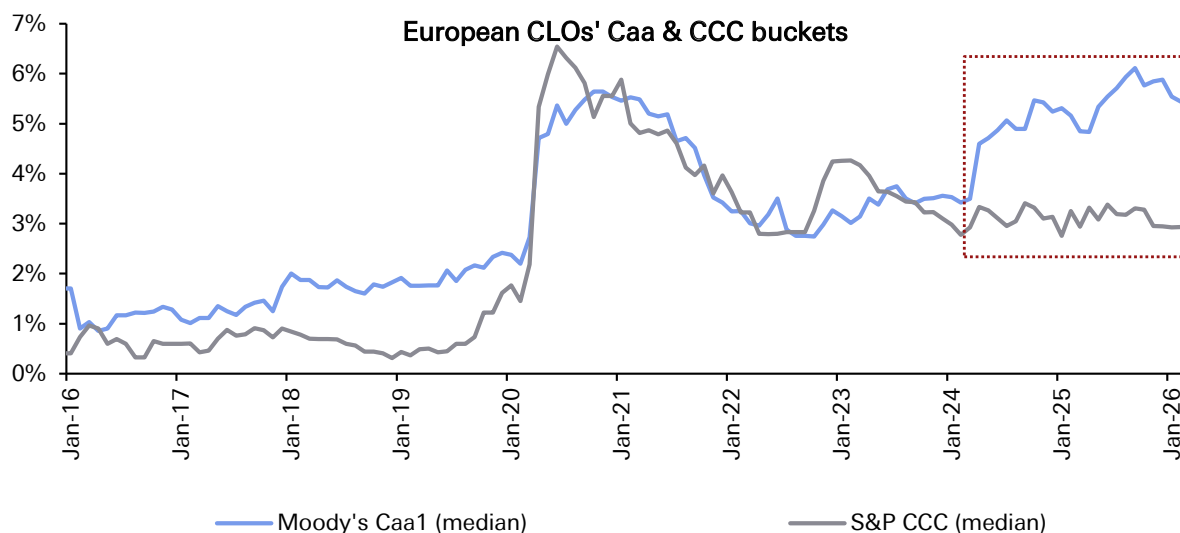


On the former, we can see from our past work on European CLO CCC buckets, that Moody's is the more conservative agency in rating spec-grade bonds & loans ([Figure 14](#)). The median European CLO has a CCC bucket of 5.4% as rated by Moody's, but only 2.9% as rated by S&P. This reflects the fact that Moody's often rates securities on an expected loss basis (taking into account both the probability of default and future recovery rates), while S&P rates securities more on a probability of default basis.

In addition, the prevalence of distressed exchanges is driving a wedge between Moody's and S&P default as well. Moody's will often classify any distressed exchange as a default if 1) the lender takes a haircut & 2) the intent of the distressed exchange is to help the borrower avoid a future default. S&P by contrast will offer more flexibility, only classifying an exchange as a default if the haircut is sufficiently large & if a default would occur immediately, without the execution of the exchange.

Lastly, there is a simple timing element as well. Moody's will classify a distressed exchange as a default effectively from the announcement date, while S&P will only shift a security into default status after the legal exchange takes place (often 1-2 months later).

Figure 14: Moody's is generally the more conservative agency when it comes to assessing ratings & default risk in Europe, and one can see this starkly in European CLO CCC buckets today.



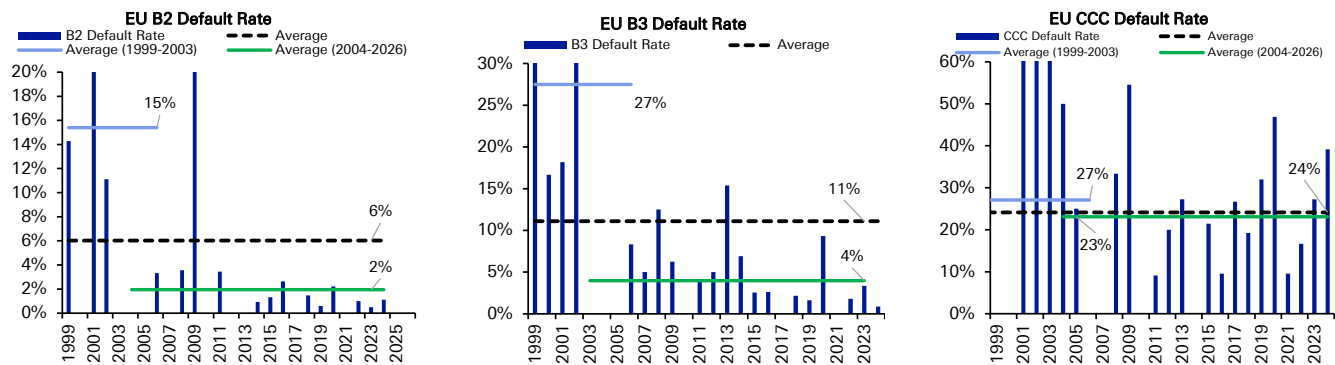
Source : Deutsche Bank, Intex, Pitchbook

There is no one answer as to which rating agency is more or less right from a conceptual standpoint. If one believes in a more benign economic scenario of low default rates and where distressed exchanges solve the problem of over-leveraged capital structures, then S&P will provide a better gauge of future credit risk. However, we have sympathy with the more conservative stance from Moody's given [Figure 6](#) already showing very weak recovery rates, even in a non-recessionary environment, coupled with the fact that many distressed exchanges are re-defaulting in future years.



Removing ourselves from the question of which rating agency has the better grip on European credit risk, we can also make some more general comments on trends in European defaults. In a similar vein to the US market, almost all of the default pressure is being generated by European CCCs, where the patience for overleveraged capital structures in challenged industries is minimal (Figure 15). ECB rate cuts last year did help to limit the damage in Bs, but this will be called further into question as the ECB looks set to hike at least twice over the rest of 2026, thanks to the fallout of the Iran war and higher energy costs.

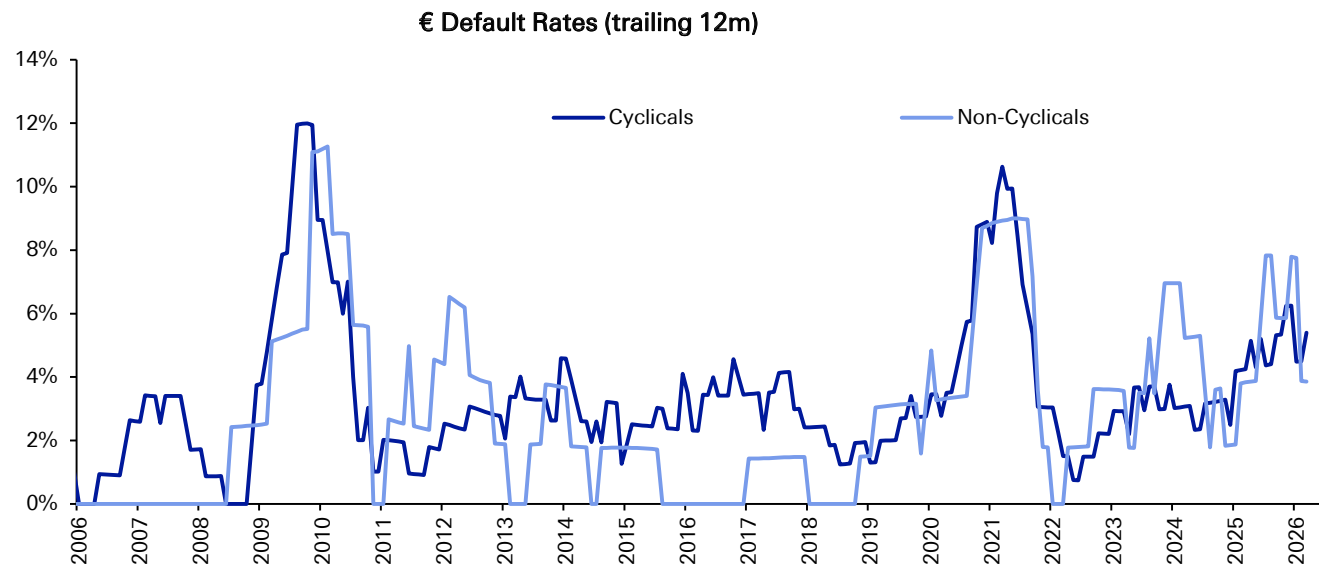
Figure 15: ECB rate cuts helped to keep B default rates limited. But even more starkly than in the US, European CCC default rates surged to above average levels. There is little patience in Europe for issuers who run into financial difficulty.



Source : Deutsche Bank, S&P

In addition, European cyclical risks remain elevated, and this is showing up in our data (Figure 16). While European non-cyclical default rates (3.9%) are similar to those of the US (3.9%), European cyclical default rates (5.4%) are now exceeding US cyclical default rates (2.1%) by a healthy margin.

Figure 16: Unlike in the US, European cyclical default rates are starting to rise

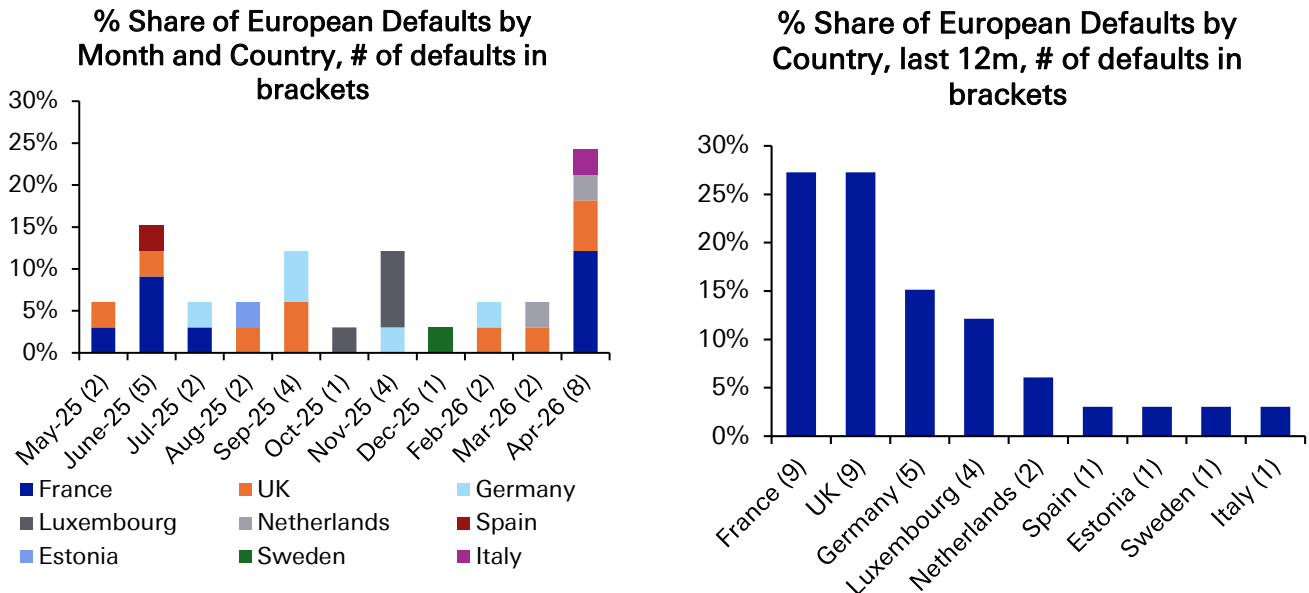


Source : Deutsche Bank, S&P



Finally, the weakest links of Europe from a country perspective are also showing up in our default rate ([Figure 17](#)). Over the last year, we have seen 9 defaults apiece from both France & the UK, 5 defaults from Germany, but only 1 apiece from Italy & Spain. This aligns with our prior work on rising bank NPLs in France & Germany, with still rapidly falling bank NPLs in Spain & Italy.

Figure 17: European defaults picked up in April, with most of the defaults over the last year coming from France, the UK & Germany.



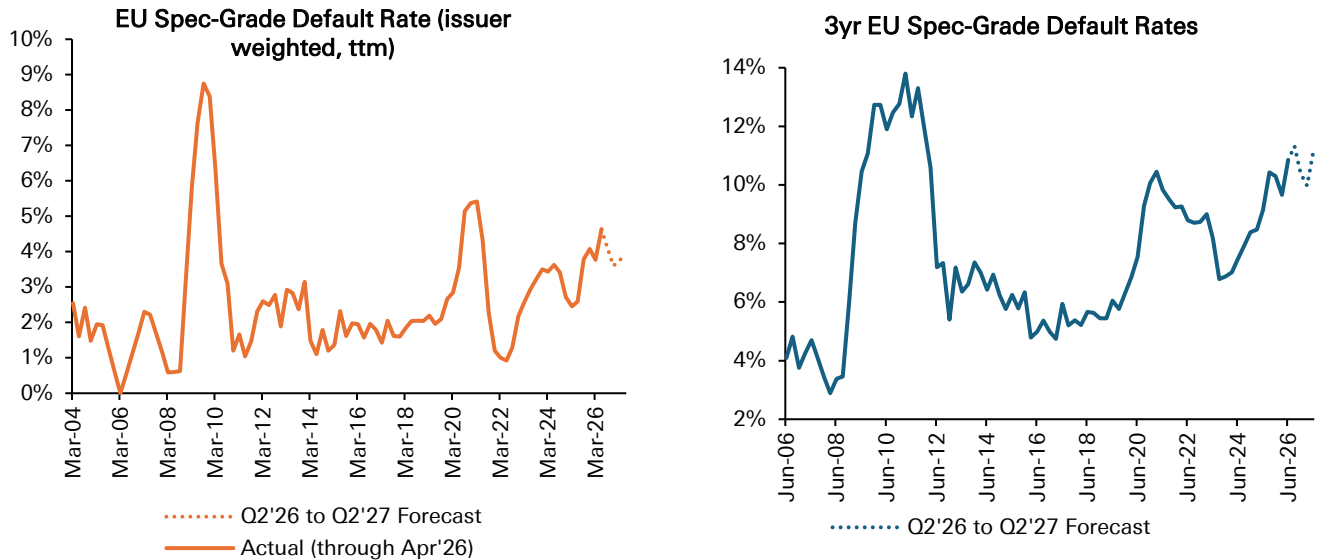
Source : Deutsche Bank, Moody's

What is the future outlook for European default rates?

A more measured pace of ECB rate hikes (June, then September) coupled with strength in Southern Europe should prevent a major spike in credit risk through the 1H'27, but default rates will remain above average, with the Iran being a major catalyst for this view. We forecast European spec-grade defaults to decline from April's very high 4.6%, but to settle at 3.6% by YE'26 & 3.9% by Q2'27. This would keep the cumulative 3 year European corporate default rates at very high levels historically ([Figure 18](#)).



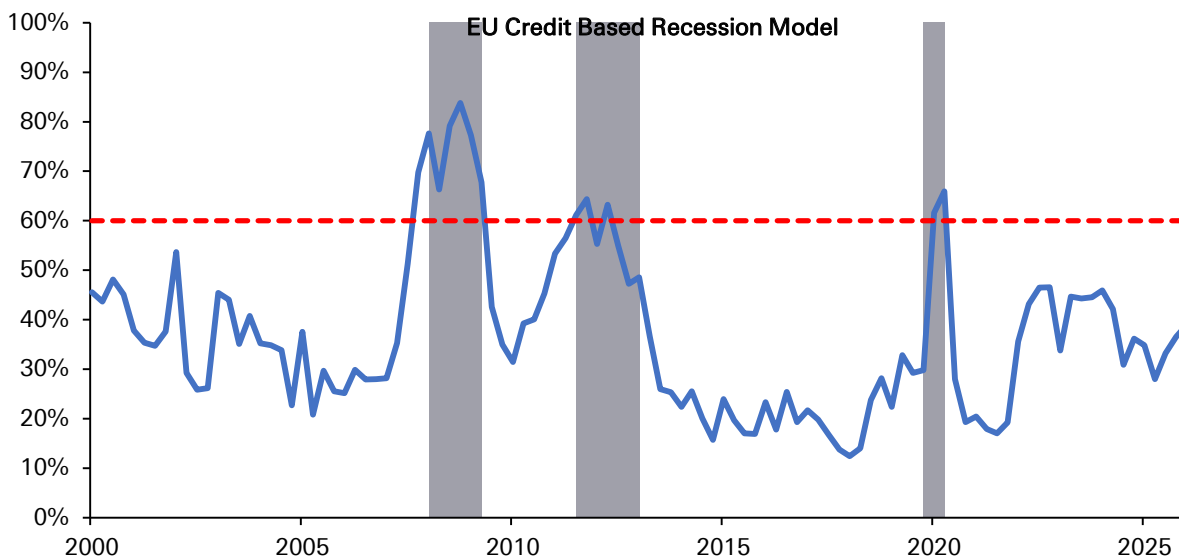
Figure 18: We forecast European spec-grade defaults to decline from April's very high 4.6%, but to settle at 3.6% by YE'26 & 3.9% by Q2'27. 3yr cumulative default rates will stay elevated.



Source : Deutsche Bank, Bloomberg Finance LP, S&P

What is the reason for our view? Firstly, our in-house European credit based recession model is far from signalling an imminent downturn, but risks are rising and are now about 0.2 standard deviations above average, with more post-war data still to be captured from our model framework (Figure 19). Unlike the US, Europe does not have an AI tailwind boosting corporate profits. And in the critical bank transmission channel, European bank lending standards are now projected to be far tighter than what is taking place in the US (Figure 20).

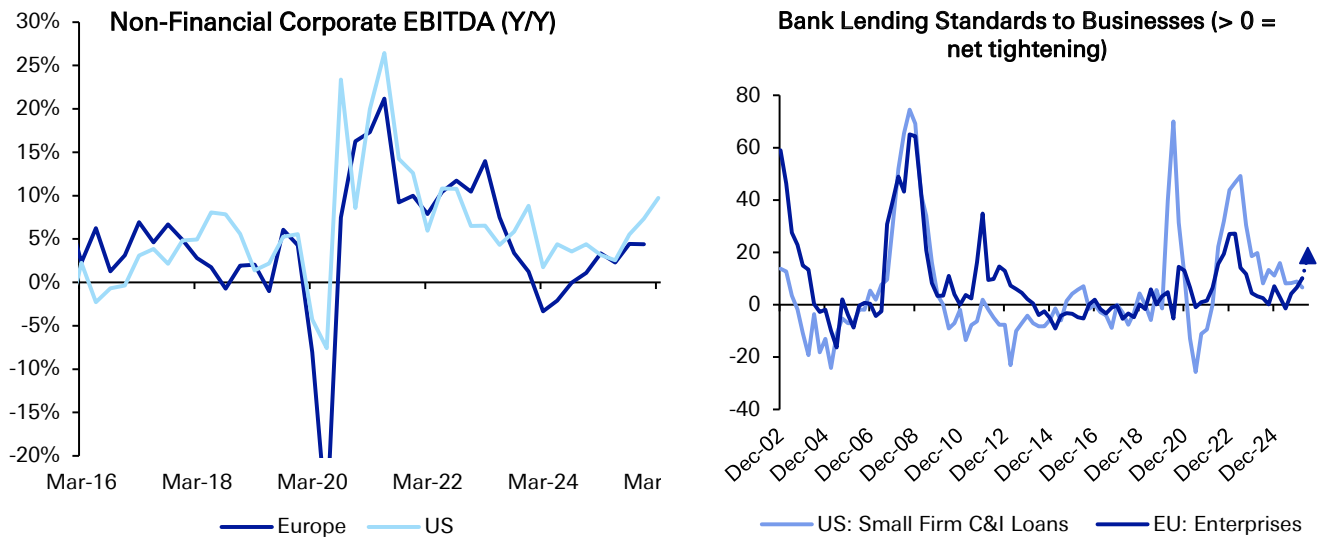
Figure 19: European cycle risks aren't alarming, but are now somewhat above average.



Source : Deutsche Bank, Bloomberg Finance LP, Haver, *Our US & EU credit indicators take an average of 0-12m forward recession probabilities across the following 4 indicators: 1) Y-Y change in GDP-based corporate leverage, 2) Y-Y change in GDP-based corporate interest coverage, 3) Y-Y change in US bank loan 30-89 day past due rates or Eurozone bank NPL rates, 4) SLOS Survey: % of banks tightening standards on small firm C&I loans or ECB BLS survey: % of banks tightening standards across business/mortgage/consumer loans.



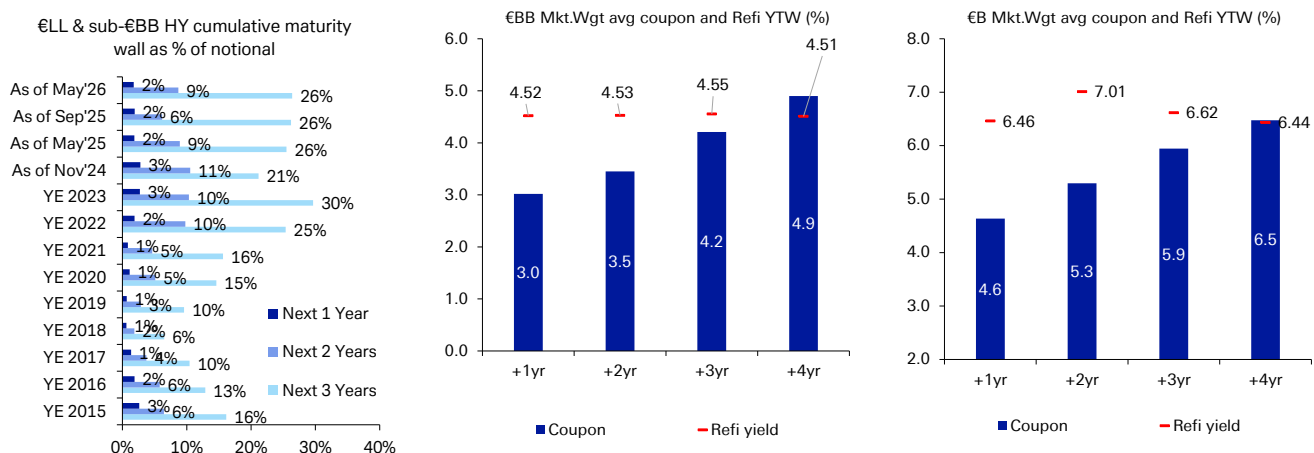
Figure 20: European corporate profits may be stalling out without the direct AI tailwinds being generated in the US. Meanwhile, European bank lending standards are now projected to tighten aggressively.



Source : Deutsche Bank, Haver, Bloomberg Finance LP

In addition, the recent shift in market expectation from ECB rate cuts to rate hikes is likely being underappreciated in terms of a market impact. We already know the maturity wall for lower-rated European spec-grade credits remains historically elevated over the next 2-3 years, especially when compared with pre-COVID levels. But in addition, B rated European companies on average would face a 70 to 170bps rise in funding costs, if these companies needed to refinance their maturing bonds in 2 to 3 years, at current market yields (Figure 21).

Figure 21: Leveraged European corporate maturity walls are far more elevated than pre-COVID levels. And the extra cost to refinance over the next 2 to 3 years will be material if the ECB isn't cutting anytime soon.



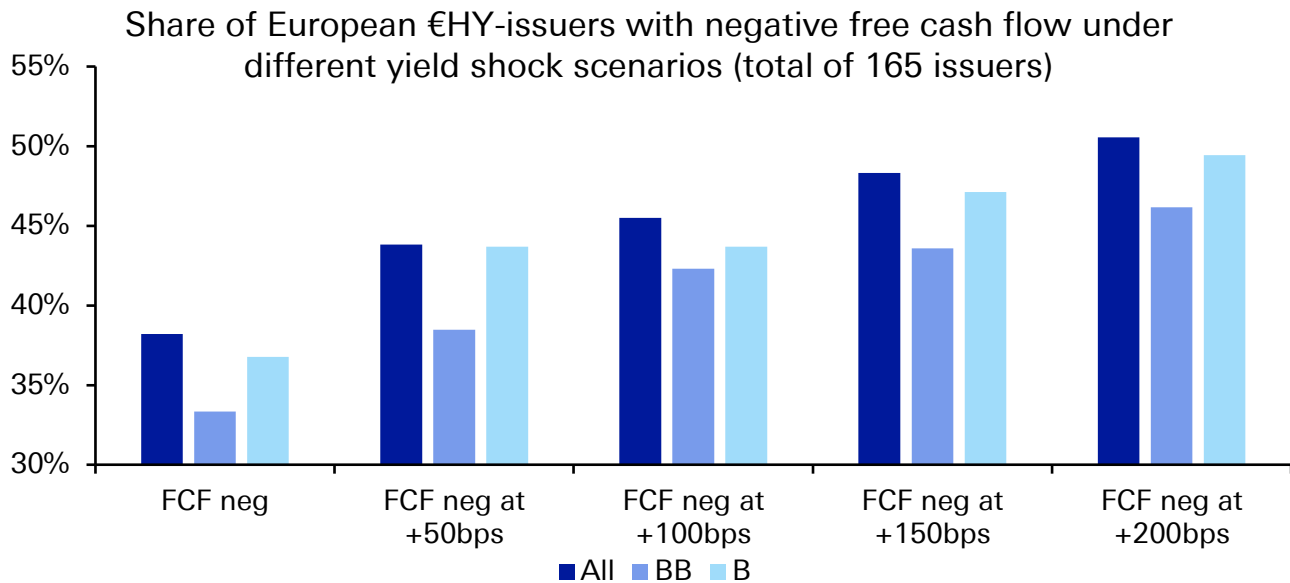
Source : Deutsche Bank, Pitchbook, Bloomberg Finance LP

This increase in funding costs is unlikely to be easily absorbed by lower-rated European HY firms. As Figure 22 illustrates below, we already face a scenario where 37% of B rated European HY issuers are currently in negative free cash flow status.



A simple 50bp increase in total funding costs would drive those percentages higher to 44%, and that is actually the biggest delta in our scenario analysis below, although clearly further increases in bond yields would drive the level of negative free cash flow to more concerning levels.

Figure 22: A simple 50bp shock in total funding costs could mean that around 45% of European B HY issuers are into negative free cash flow status



Source : Deutsche Bank, Cognitive Credit
Only Issuers from European Countries

Put simply, a small increase in funding costs can drive meaningful differences in European cash flow generation. And this is one reason we have become more bearish on Europe in recent months. Weak growth is not the sole factor of why our stance here has shifted. European spec-grade markets could face moderately weaker growth without higher defaults, with the ECB acting as a relative tailwind through insurance rate cuts. But as it stands today, the Iran war is likely to leave Europe with both growth and central bank headwinds at the same time.



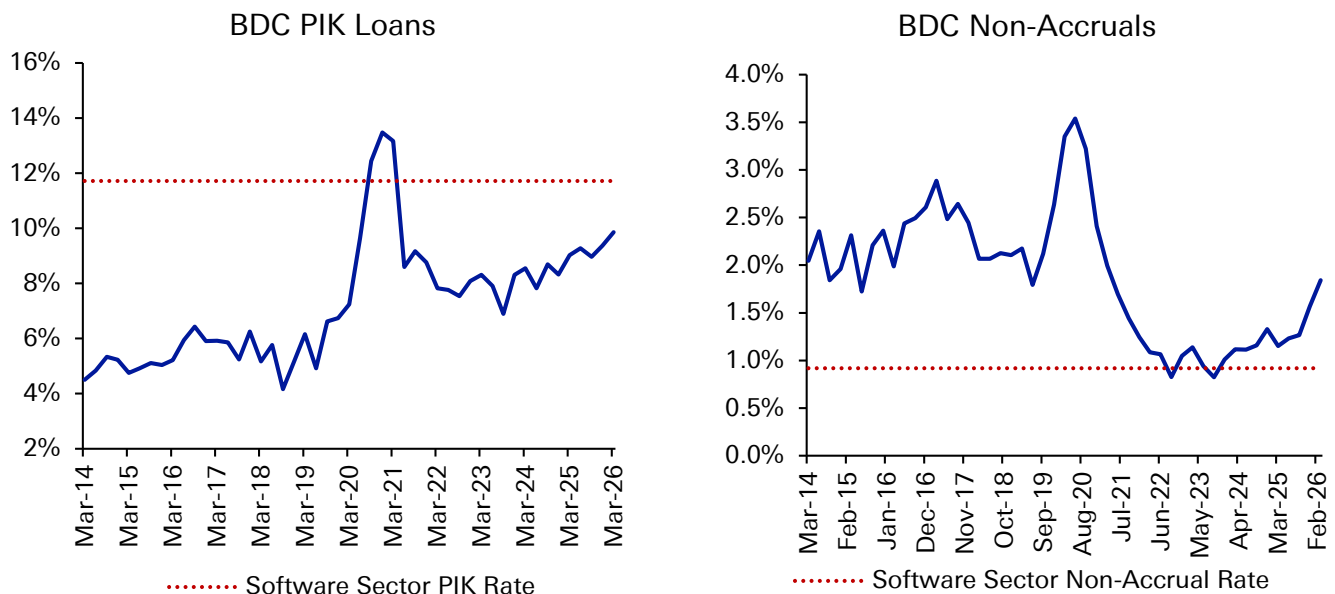
Tactical and structural factors to watch

A US Software glitch can materially tighten funding markets

As discussed in our abstract, the tailwinds of AI disruption for aggregate growth will likely become headwinds for spec-grade credit, through tighter financial conditions in private credit & leveraged loans. This will eventually play itself out through two channels. The first will be more hawkish Fed policy, but the second will be meaningful downgrades of US software names into CCC buckets, where investors have already demonstrated their impatience to provide flexibility to this rating cohort over the last few years. In essence, risks are rising that we could see a scenario in early 2027, where tangible AI disruption of software business models begins to damage software revenue and customer retention rates, at the same time actual Fed hikes eat up limited software free cash flow.

The implications of this scenario are not being priced. In private credit specifically, PIK interest usage (paying interest in kind, rather than in actual cash) is elevated both in aggregate and for software names, but non-accrual rates are still below averages, with plenty of room for software non-accrual rates to rise over the next few years to catch investors off-guard ([Figure 23](#)).

Figure 23: Private credit PIK rates and non-accrual rates are rising. But while software PIKs are high, software non-accruals are still well below average, and this will likely change over the next 1-2 years.



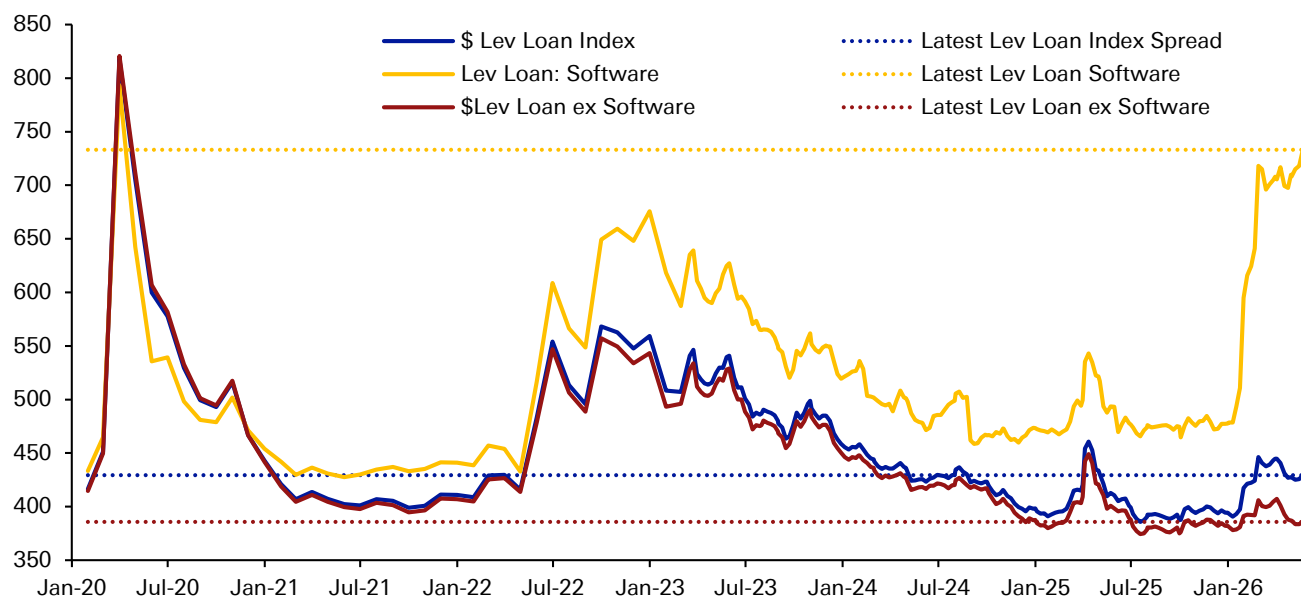
Source : Deutsche Bank, SEC EDGAR

And in the leveraged loan market, software risk has rapidly re-priced, but with an uncanny lack of spillovers to other parts of the market, in a way that makes the divergence between \$HY energy and ex-energy in 2015 look tame ([Figure 24](#)). US Leveraged Loan software spreads are now trading at 733bps, at YTD wides, and



trading at the widest levels since March 2020. However, software stress remains isolated, with ex-software spreads trading at 386bps, below 2021 levels, when central banks were still running historically easy monetary policy. This is an astonishing divergence between software & ex-software leveraged loan spreads, especially considering US software is the largest sector in both the US Loan and Private Credit markets. Future software stress, most tangibly in the form of rating downgrades, would cascade through broader spec-grade financial markets in a more disruptive fashion than is appreciated.

Figure 24: Software glitch? US Lev Loan software spreads are trading at 733bps, near the YTD wides



Source : Deutsche Bank, Pitchbook

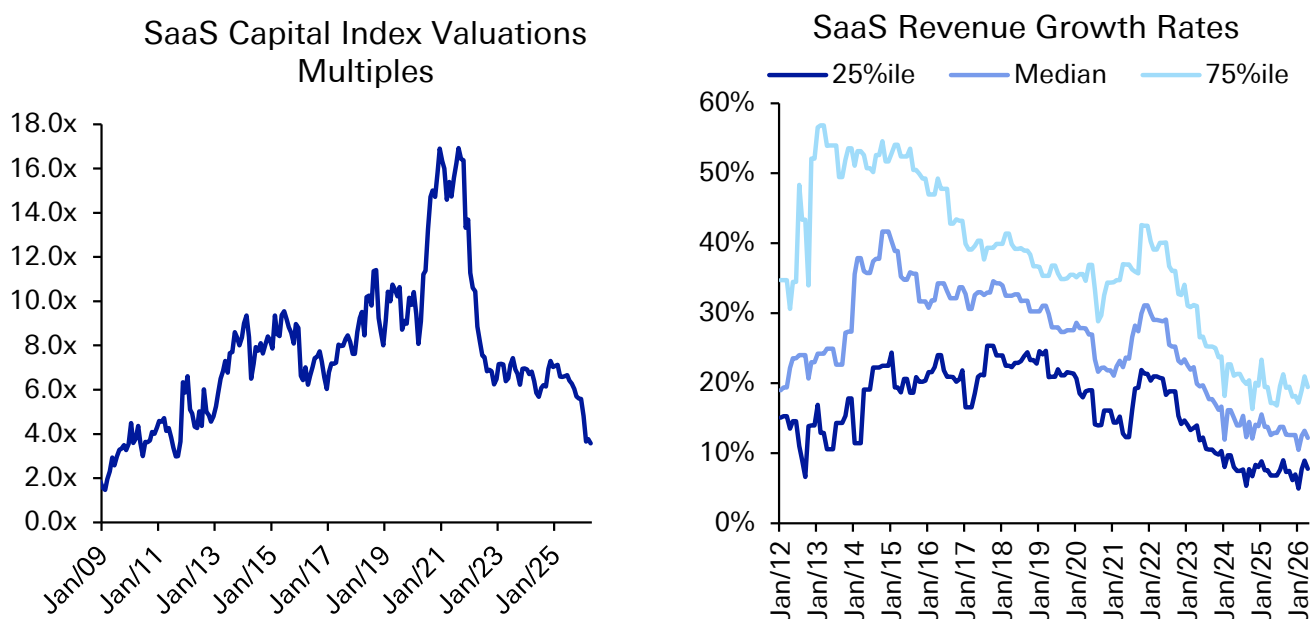
It is clear loan investors remain concerned about the impact of AI disruption on the long-term viability of the software sector. AI disruption to Software comes in many flavors. First, it reduces clarity on future recurring revenue growth, as customers may decide to forgo renewing contracts. This is critical for the leveraged software space, since a large share of firms currently have negative EBITDA.

Second, AI will likely reduce white-collar employment over time, which cuts against the “per seat” model that many software firms employ to earn revenue. Third, it paradoxically could encourage many software companies today to over-invest in AI (“if you can’t beat ‘em, join ‘em”), before knowing how successful AI models will eventually become, or how best to integrate them. This isn’t a trivial concern, given that many SaaS firms have limited financial flexibility. An investment mistake today could prove costly later, if financing markets seize up.

Put simply, equity & credit investors are making a judgement call that a universe of highly leveraged B2 & B3 rated names, facing rather large maturity walls over the next 2-3 years (which need to be refinanced within the next 1-2 years), will have difficulty adapting to a world of increased AI adoption. And this is why software multiples to annualized recurring revenues have dropped nearly 50% since 2024, implying that LTVs for leveraged software firms are far higher than the 40-50% often cited by investors in the space ([Figure 25](#)).



Figure 25: Software multiples have collapsed 50% since 2024, implying LTVs are far higher than advertised. But the biggest catalyst for further software weakness will be when revenue growth is disrupted.



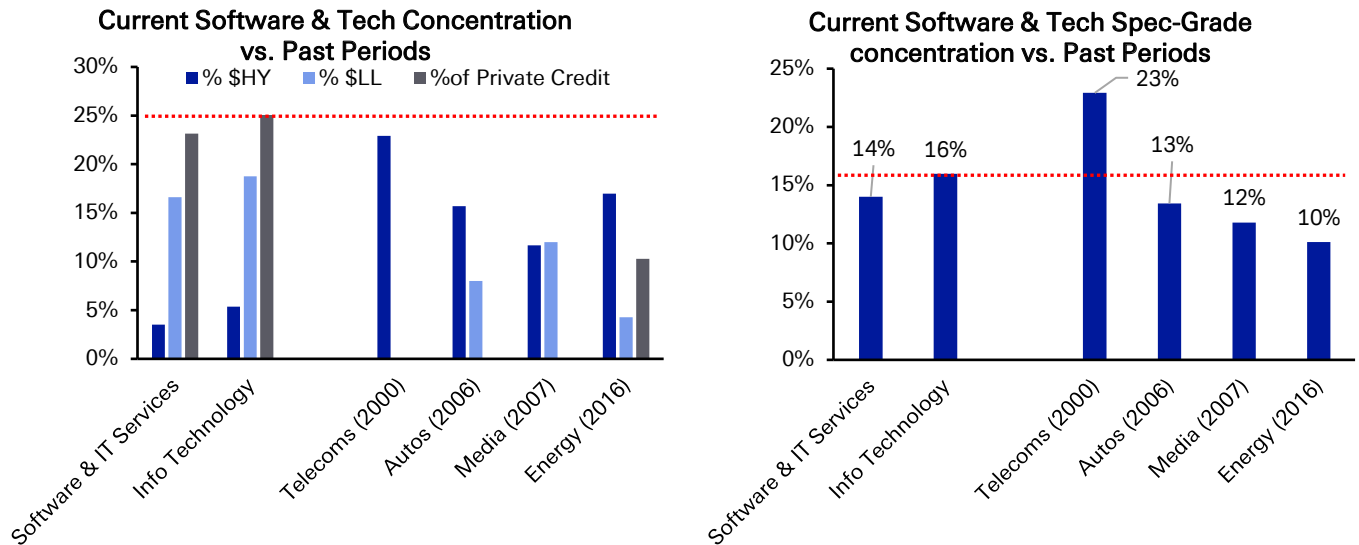
Source : Deutsche Bank, SaaS Capital Index

The challenge of course is timing: When will this AI disruption will actually show up in software company fundamentals? To-date, one can see in Figure 25 that while software multiples have collapsed, software revenue growth rates have remained stable. This is one reason why rating agencies and private credit funds (such as BDCs) are not reacting to the large drop in secondary prices within the leveraged loan market. The answer on the timing question will determine whether software modestly or materially tightens US spec-grade financing conditions over the next year.

But structurally, it is useful to consider the transmission channel of how software stress would spread in a manner that could matter meaningfully to both credit & multi-asset investors. The first point is that software is a very large and consequential sector in historical context. Software comprises a larger share of US spec-grade credit today than Shale Energy did in 2016. And in private credit specifically, software & software-related names comprise a greater share of the asset class than the Telecom sector did to \$HY in 2000 ([Figure 26](#)).



Figure 26: The Software and Technology exposure within US Private Credit is larger than Telecom's exposure within \$HY in 2000. Across all of \$-spec-grade credit (HY, LL, Private Credit), software & tech exposure will be more impactful than Shale Energy in 2016



Source : Deutsche Bank, Bloomberg Finance LP, Pitchbook
*Private Credit estimate is based on projected investable assets relative to observable BDC sector concentrations

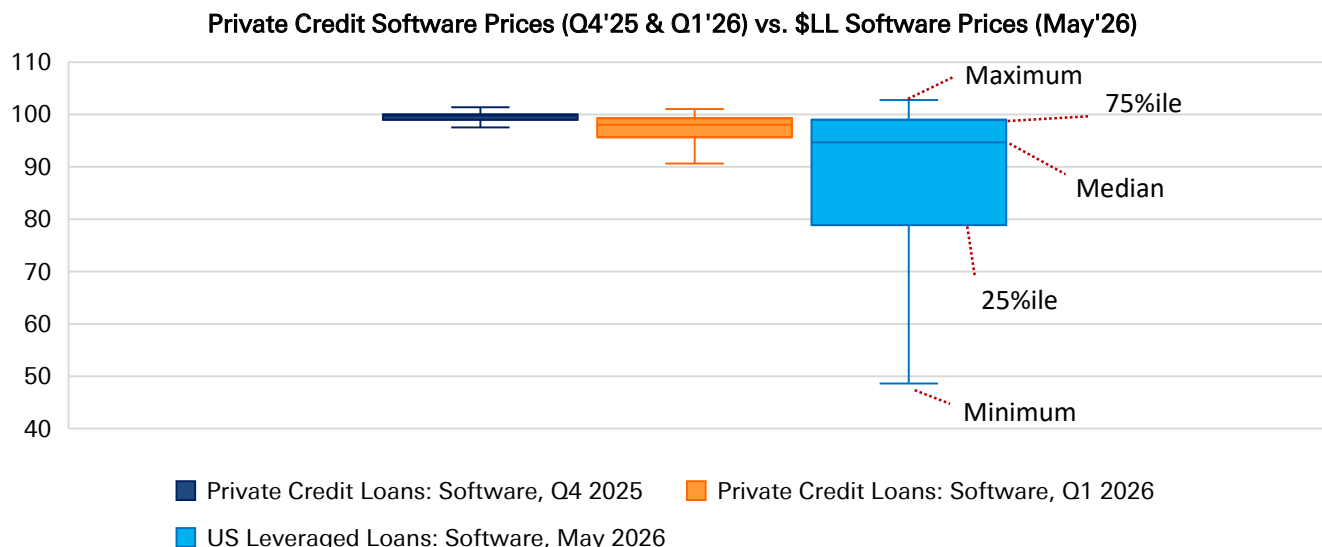
But there is still inertia in the overall credit ecosystem that is preventing this software stress from propagating more broadly. To us, this inertia cannot last forever.

The first unstable equilibrium can be found in US private BDCs, where still stale software loan markets are leading to liquidity pressures and storing up future problems for the asset class. We can first see this within US private BDCs ([Figure 27](#)), where we analyse the 7 largest private & "semi-liquid" BDCs in the market today. After combing through their 10-K & 10-Q filings, we see that any software loan markdowns taking place are very gradual and ultimately insufficient ([Figure 27](#)).

For example, through the end of Q1'26, the median private credit software loan was still valued at a price of 98, while the 25th percentile of private credit software loans was still valued at 95.6. This is far more optimistic than what the very similar US leveraged loan market is signalling. The median US Lev Loan software name trades at 95.2, while the 25th percentile of US Lev Loan software names trades at 78.9.



Figure 27. The private credit market is pricing that AI will only have a minor impact on leveraged software names. The reality is more likely priced in Leveraged Loans, where the impact is more dispersed, but far more material*



Source : Deutsche Bank, Bloomberg Finance LP, SEC EDGAR Form 10-Ks

*Private Credit Software loans were pulled from the 10-K and 10-Q filings of the following 7 private BDCs. Private Credit loan price is calculated as Fair Value/Par Value. US leveraged loan software prices are pulled from the Bloomberg US Leveraged Loan Index. The plots above exclude outliers that are more than $\pm 1.5x$ the interquartile range (75th - 25th percentile).

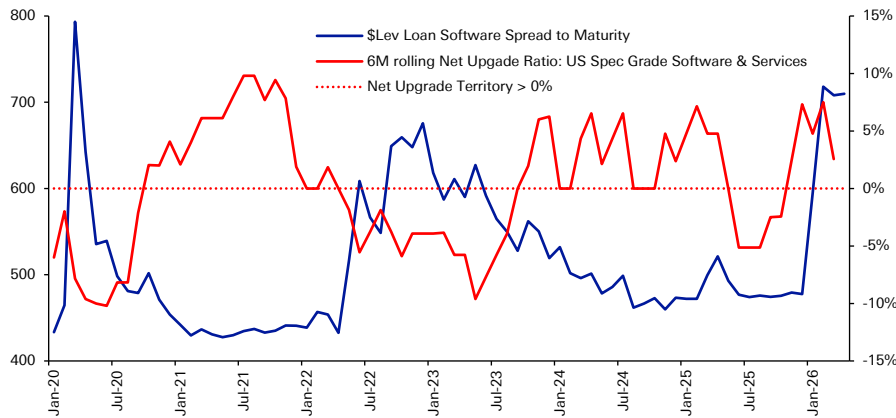
This inertia in private credit market pricing is quite striking in the tails of the distribution. Careful readers will just have noted the 25th percentile of private credit software names are marked with a higher valuation than the median US Leveraged Loan software name. And when we look at distress ratios, the story is similar. Only 5.8% of private credit software loans were valued at a price less than \$80 at the end of Q1, compared to 25.6% of US Lev Loan software loans today that are now trading below \$80.

Ultimately, US private BDCs continue to price a very tight software distribution, in essence implying that AI will have a very broad, but ultimately, minor impact on the entire software space. We suspect the reality will entail a lot more dispersion, with a much fatter left tail for private credit software companies.

But the second unstable equilibrium is coming from the rating agencies. Despite a massive widening in software spreads, US software names are still currently in net upgrade territory over the last 12 months, as proxied by S&P (Figure 28). This disconnect is clearly unsustainable. Either software spreads will tighten notably from here, or software downgrades will pick up more broadly in earnest.



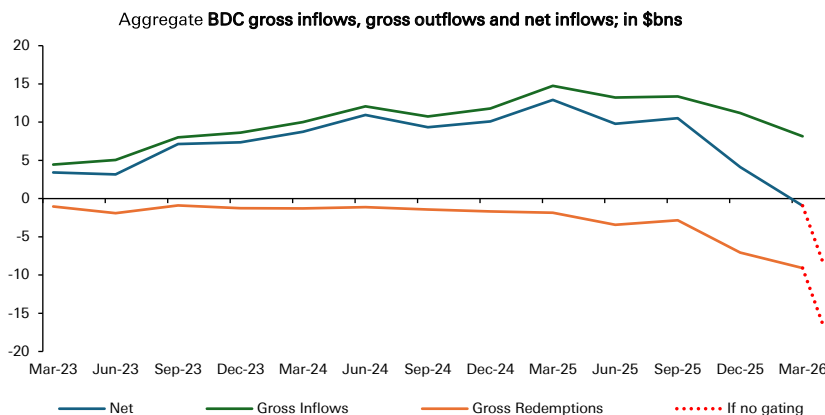
Figure 28: Software spreads have surged YTD, but rating agencies still have the spec-grade software sector in net upgrade territory. This divergence won't last for long.



Source : Deutsche Bank, Pitchbook

These unstable equilibriums will eventually prove problematic. The short-term problems from stale private credit software loan marks will continue to feed into US BDC liquidity pressures. Private BDC investors will continue to utilize their “first-mover advantage” and withdraw their capital, before private BDC software loans, and eventually NAVs, are marked lower in future quarters. We already experienced a wave of redemption requests in Q1, but we would expect new inflows to slow down dramatically from here as well, forcing BDCs to 1) increasingly issue more debt, 2) tap bank revolvers, and 3) sell leveraged loans to raise needed cash to meet growing net outflows. This part of the story is already in play, but will likely get worse in future quarters (Figure 29).

Figure 29: Continued divergence in private vs. public market pricing is likely to drive greater net outflows from BDCs in the quarters ahead, mainly as gross inflows into private BDCs now slow.



Source : Deutsche Bank, SEC EDGAR
Includes 7 private BDCs and 1 interval fund. Pulling from 10-Q and 8-K filings as of Q1 2026, and from PORT-P filings as of Q4 2025.

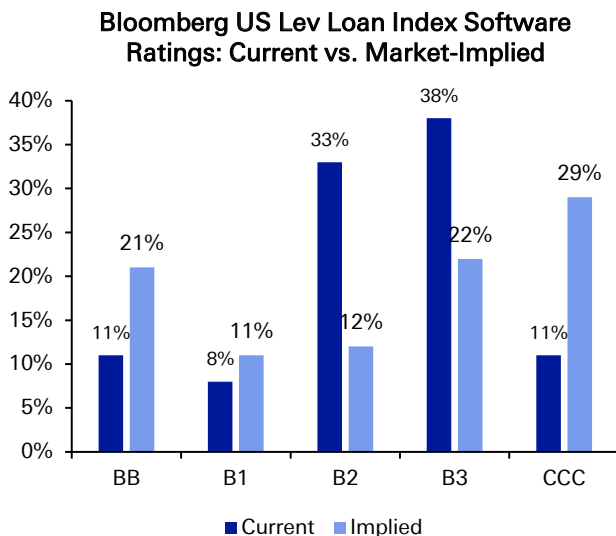
But an actual deterioration in software fundamentals is likely to be the catalyst from here that could meaningfully tighten financial conditions, as the US rating-agencies and third-party valuers assessing private credit fund portfolios quickly adjust their



models. Again, the timing here is uncertain. But actual evidence of slowing revenue growth, lower free cash flow generation, and/or reduced customer retention ratios will reduce the ability of many software companies to refinance upcoming maturity walls, certainly increase rating agency downgrades.

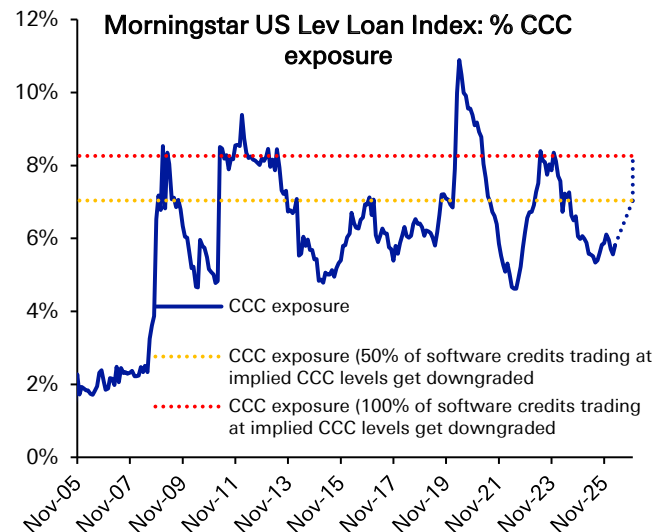
And if this fundamental pressure does emerge, the downgrade pressure on broader-spec grade credit could be significant. Based on current market pricing, we estimate that 29% of current US Leveraged Loan software names are trading with an implied CCC rating, vs. only 11% rated as CCC by the rating agencies (Figure 30). Given the large weight of software (~13%) in the US Lev Loan index, this could spike the share of CCCs in the Loan Index to 7-8.5% (from 5.8% today, Figure 31), on par with 2008-2009, 2011-2013, 2016-2017, 2020 & 2022-2023 levels (Note this 7-8.5% CCC basket is just looking at the software downgrade risk in isolation, without accounting for any additional downgrades hurting other sectors, whether from direct AI disruption of non-software sector business models to indirect AI disruption tightening financial conditions through tighter Fed policy and higher rates).

Figure 30: Market prices imply that nearly 30% of US software lev loans should be CCC-rated*



Source : Deutsche Bank, Bloomberg Finance LP, *market-implied ratings are calculated by comparing current individual \$LL software prices vs the average price per \$LL rating bucket. We assign a market implied software rating based on the average rating bucket price that is closest to the individual software loan price

Figure 31: If true, the share of CCCs in the US Lev Loan index will swell. That is the risk of the largest sector in the index facing AI disruption ahead.



Source : Deutsche Bank, Pitchbook

And from the third-party valuer who assess the veracity of private credit loan marks, it is a very similar dynamic. Third party valuers effectively use calibration models to assess the "fair-value" of a private credit loan each quarter, using both market-wide macro inputs (such as traded loan index and sector spreads) and company-specific fundamentals to inform their view.

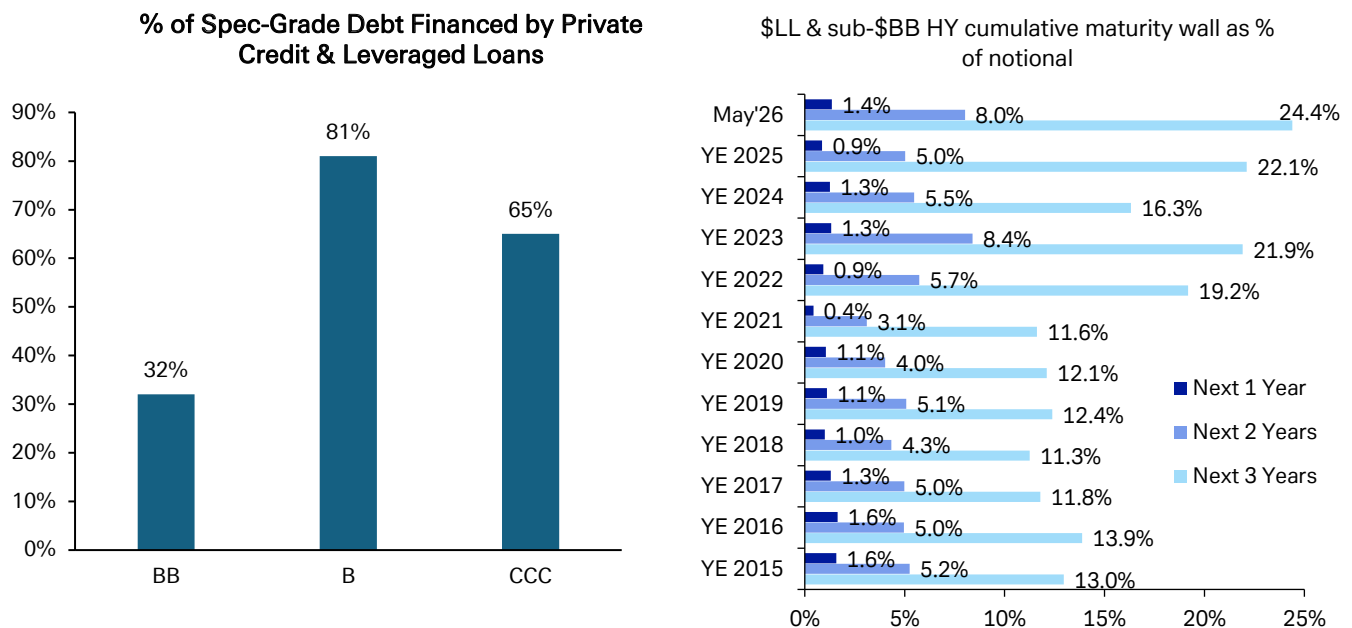
Currently, the dramatic widening in US Leveraged Loan software spreads is factoring into modest private credit loan markdowns today. But 1) the health of the broader Leveraged Loan Market (ex-software) & 2) the lack of any tangible impact on software fundamentals yet is limiting the amount of loan markdowns that



private credit funds have to take. A raft of weak software earnings coupled with rating downgrades and rising CCC buckets in US Loan indices would lead both of these macro & company specific fundamental factors to deteriorate, instigating a far more rapid pace of private credit software loan markdowns than what we have experienced to-date.

To sum it up succinctly, we suspect software stress is likely to spread more broadly, within both private credit & leveraged loan markets. With 80% of all \$B rated debt financed in either private credit or leveraged loans, the reality of still near record high T+2 & T+3 maturity walls for \$B & CCC rated credit highlights that this is a risk not to forget (Figure 32).

Figure 32: 81% of all B-rated financing in US spec-grade credit comes from floating-rate Leveraged Loans or Private Credit. A problem if these markets face more pressure, given maturity walls 2 & 3 years out are at all-time highs.



Source : Deutsche Bank, S&P Capital IQ Pro, Bloomberg Finance LP, Pitchbook

What markets/sectors are most exposed to upcoming risks?

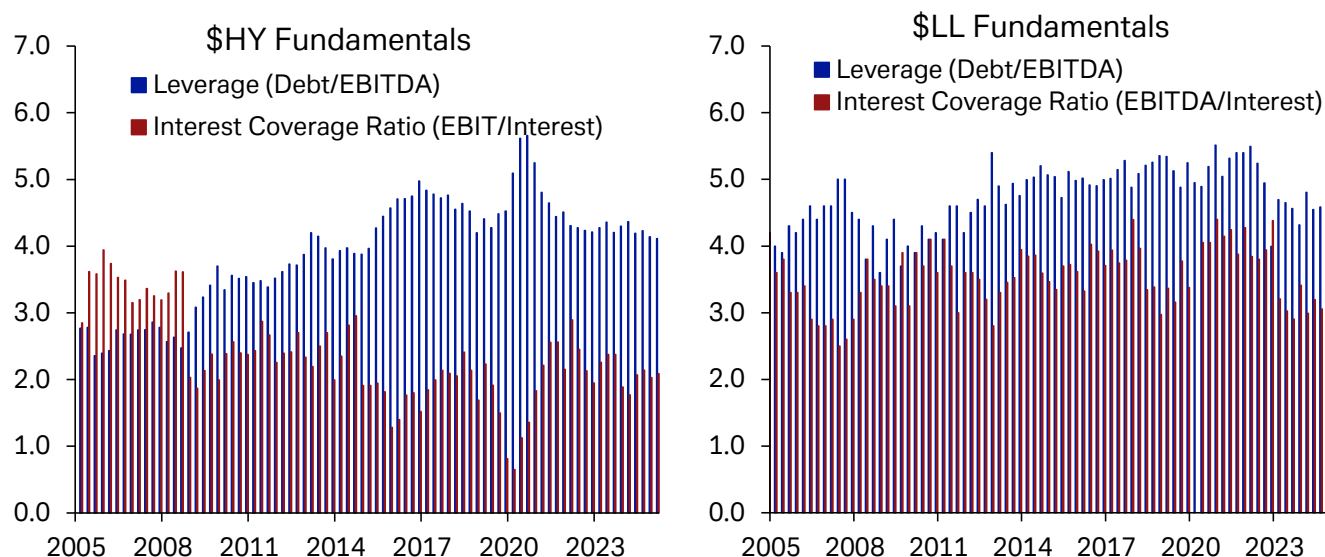
As laid out above, **there are many intertwined risks that investors have to navigate in the months ahead.** If rising inflation leads to a more hawkish Fed, then the expectation for higher-for-longer financing costs could lead multiple contraction. Higher energy and commodity costs can also compress margins as consumers have less fiscal headroom than in the recent past. While AI-led US growth has powered the US economy, unless it broadens out there could be greater earnings growth dispersion.

\$HY and \$LL leverage had been rising in the years following the Financial Crisis until the post-Covid period. Since 2021, there has been a gradual easing in leverage. However, **even as leverage has gently eased, interest coverage ratios have remained quite contracted** over the last two years. This leaves spec-grade credit quite sensitive to higher rates moving forward. This is not a perfect comparison between markets given data constraints, but the trends are informative. The \$HY



data is the median issuer per quarter, while the Lev Loan data is an aggregation at the index level for new issuance.

Figure 33: USD HY and LL leverage continues to ease from Covid-era highs, but interest coverage ratios have not improved in a similar fashion



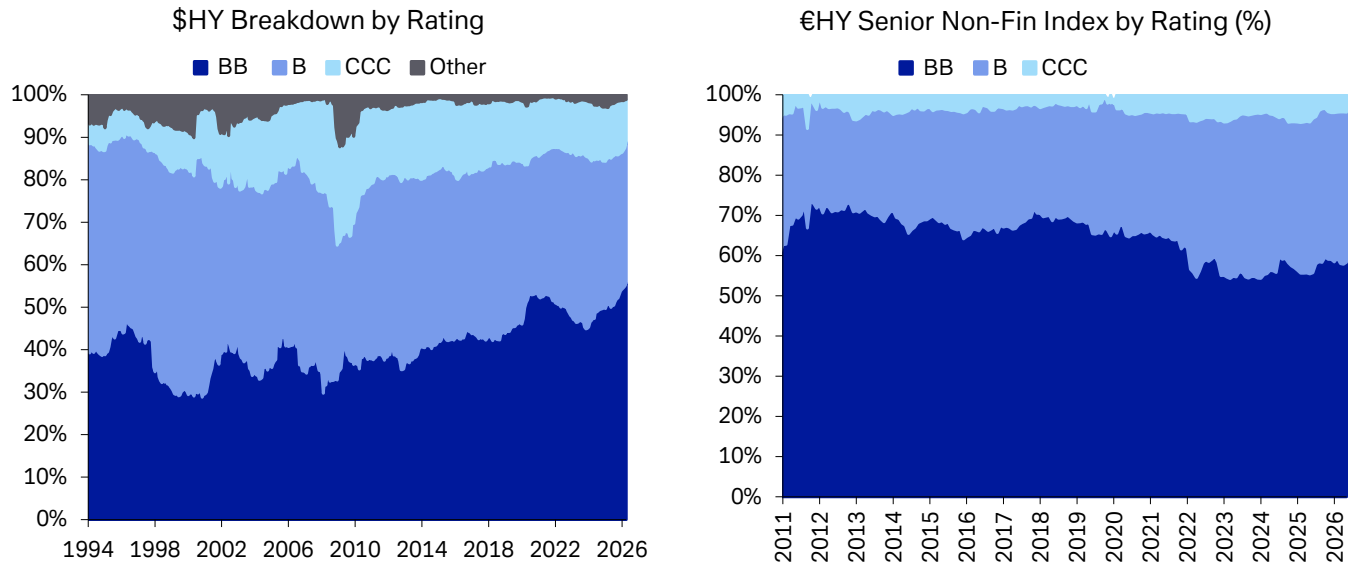
Source : Deutsche Bank, Bloomberg Finance L.P., Pitchbook LCD
\$HY data uses the median issuer by quarter

Fixed-rate High Yield and floating-rate Leveraged Loan markets continue to grow further apart in average rating. \$HY markets have further improved in rating quality over the last year. **\$BBs are a record share of the \$HY index at 56%**, while \$CCCs now only make up 10% of the index. That is the lowest share of \$CCC bonds since 2002. The improvement in \$HY rating quality is due to multiple factors. One is that the HY market is looking to finance higher quality issuers. There is also a greater sum of subordinated-debt from \$IG rated issuers. And lastly, there has been many recent large downgrades over the last two years that have introduced significant \$BB debt into \$HY.

€HY rating quality is slowly going the other way. Around 60% of Non-Financial seniors are rated €BB, which is down nearly 10pp from the highs in 2018. €CCCs still represent a smaller portion of this market with just 4% of notional value. As the level of €B debt grows, it introduces the potential for European defaults to move slightly higher structurally.



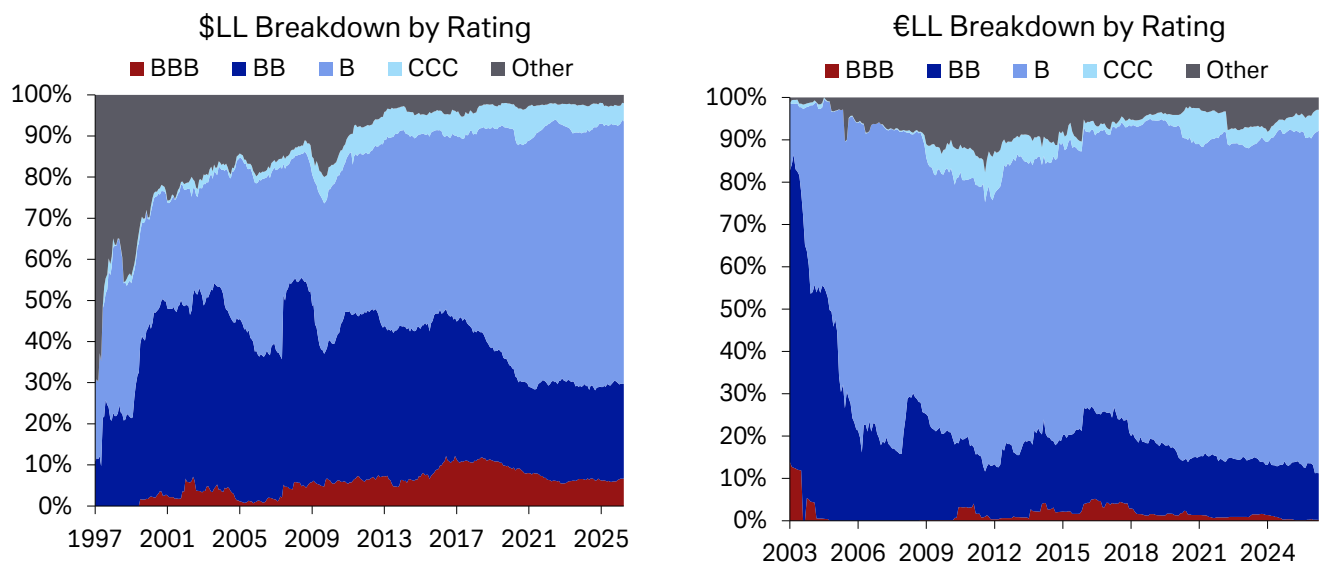
Figure 34: While \$HY bond markets have steadily increased in credit quality over time, €Bs have become a more substantial part of €HY.



Source : Deutsche Bank, Bloomberg Finance L.P., ICE Indices, S&P

\$ & € Leveraged Loan markets have become largely dominated by B-rated issuers. The share of \$B Loans is up to 64% from 45% in 2015, having been just shy of two-thirds of the index since 2022. Meanwhile €B Loans are a record 81% of the market after being 66% of the market at the end of 2015. This shift of rating quality lower in loans relative to HY partly explains the divergent default rates between the two markets during the post-Covid cycle.

Figure 35: While Leveraged Loan indices have long been overweight B-rated debt, we are at a record share in EUR LL and just off the largest concentration in USD LL.

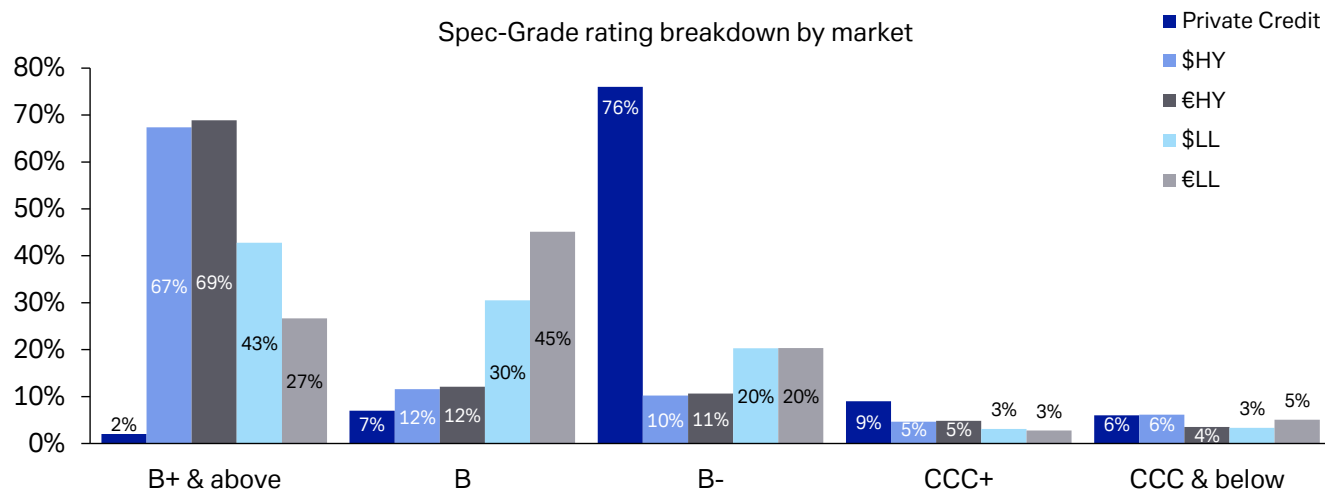


Source : Deutsche Bank, Bloomberg Finance L.P., ICE Indices



Looking ahead, **Private Credit is even more exposed** to acute risks. Both \$ & € HY markets are over two-thirds rated B1 and above, with just over 20% B2 rated risk, and just around 10% of CCC-rated securities. \$LL is better rated than €LL, with a plurality of debt rated B1 or better, while the largest share of €LL debt is rated B2. By contrast, **Private Credit is over 75% B3 rated**, just one notch away from a CCC-rating. Speaking of which, **Private Credit also has 15% CCC or worse rated debt**, while \$HY markets is the closest public market with 11% of notional marked below a B3-rating.

Figure 36: Private Credit is lower rated than Lev Loan indices with an extreme concentration of B3 rated debt



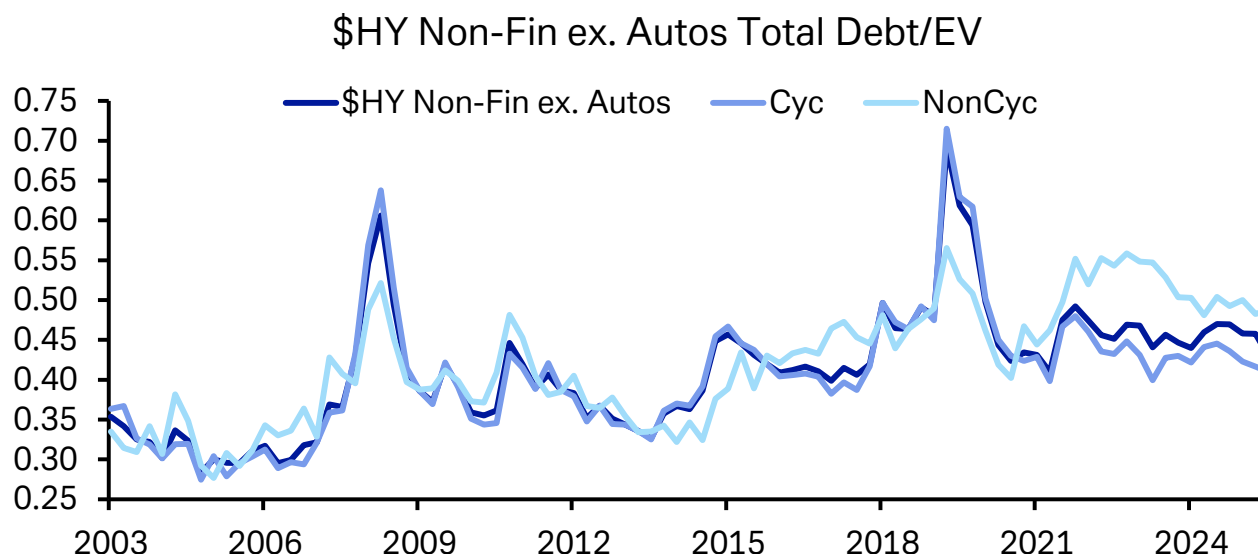
Source : Deutsche Bank, Bloomberg Finance L.P., Pitchbook, S&P Cap IQ

Which sectors are more exposed to defaults and fundamental risks?

We have found that debt to enterprise value ratios provide a better leading indicator of future downgrade and default rates than other leverage measures such as debt to EBITDA. This is based on a Merton Model framework, where falling equity valuations can create incentives to restructure the firm's capital structure, often to the detriment of credit holders. **Debt to EV ratios at the index level have been improving marginally over the last two quarters.** While the median \$HY member may not necessarily be seeing the same multiple expansion that is evident in the S&P 500, \$HY valuations have still been buoyant. Meanwhile debt levels have not risen substantially over the last few years. As such Debt/EV ratios are near the bottom of their range since the end of 2021, with Cyclical firms seeing more improvement than Non-Cyclicals.



Figure 37: The recent risk rally has caused \$HY cyclical leverage to move toward post-Covid lows, even as Non-Cyclical leverage remains steady over the last year



Source : Deutsche Bank, S&P, Bloomberg Finance LP, ICE indices

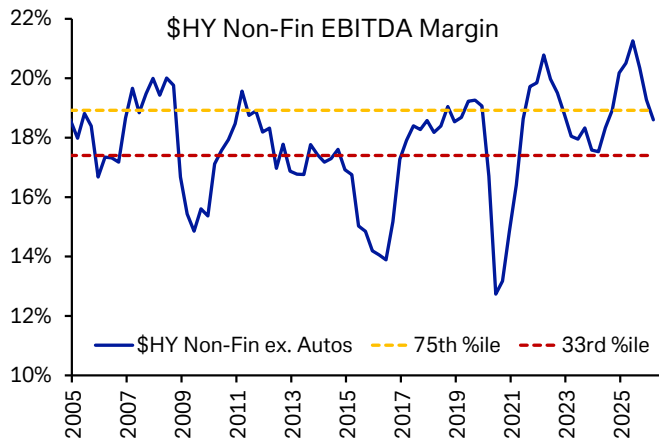
Looking ahead to the next 12 month, **multiple risks can drive greater dispersion**. The energy shock and rising inflation worries are translating into higher US Treasury yields which can lead frothy multiples to re-rate lower. Rising energy and goods costs can also weigh on margins, which could lead to higher leverage for Spec-grade corporates in the coming quarters.

In order to model how these factors can effect leverage across sectors, we have created a scenario analysis in [Figure 40](#) below. **The first scenario is one in which the Strait of Hormuz opens up shortly** and there is a smaller commodity shock, keeping margin compression moderate. We also would likely see less inflationary pressures under this scenario and therefore less multiple compression as the Fed does not grow as hawkish as they might otherwise. We grow current revenue by the median 12-month revenue expectation in each sector, and then estimate EBITDA growth based on the 75th %-ile observed profit margin since 2004. We then re-rate the sector at the 75th %-ile EV/EBITDA. If current multiples or margins are lower than the 75th %-ile historically, we just use the current levels.

In the second scenario, there is a more sustained commodity shock, wherein oil prices are higher-for-longer, likely leading to higher yields on the back of higher inflation expectations and rate hikes. We still use the current revenue growth forecast but then contract multiples to their long-term median and margins to their 33rd %-ile unless they are already lower. For context, [Figures 38 & 39](#) shows what these levels look like relative to historic \$HY equity multiples and EBITDA margins.

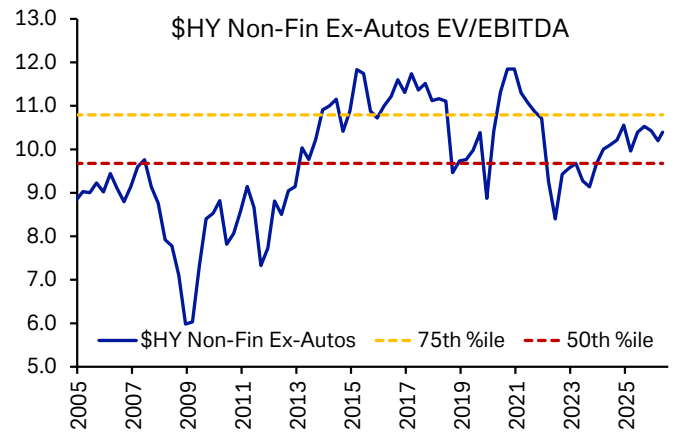


Figure 38: \$HY EBITDA Margins have compressed over the last 2 quarters before any increase from the current commodity shock



Source : Deutsche Bank, S&P, Bloomberg Finance LP, ICE indices

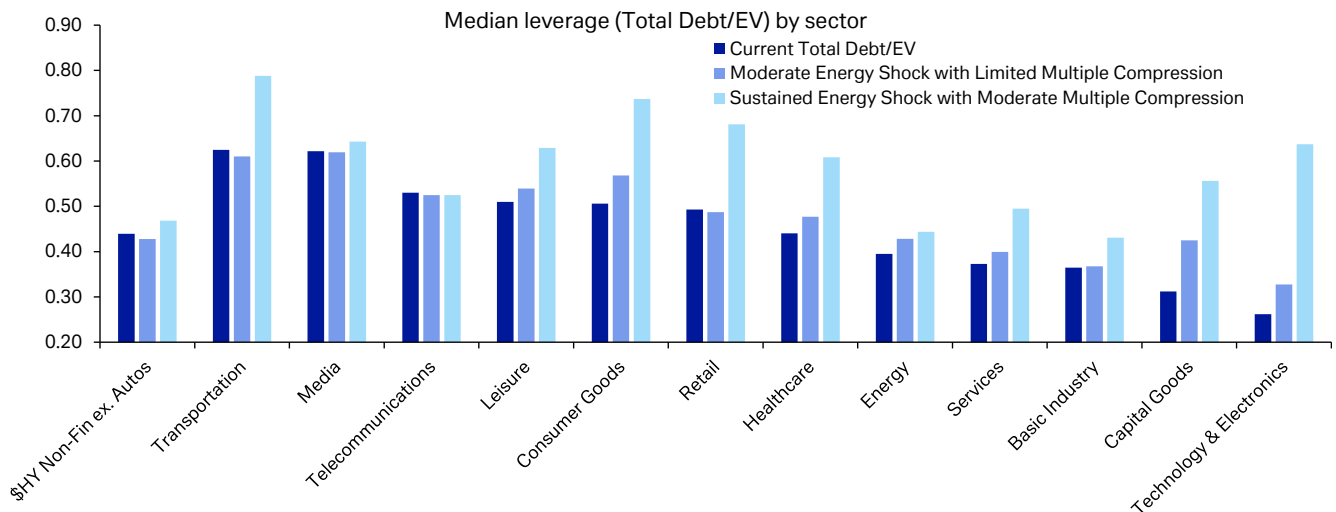
Figure 39: Equity multiples of the median \$HY issuer are near the 75th %-ile historically, and had moved toward the median level during recent hiking cycles



Source : Deutsche Bank, S&P, Bloomberg Finance LP, ICE indices

As we laid out above, \$HY Cyclical sectors are starting out with better leverage metrics while some Non-Cyclical sectors have higher outright leverage or high leverage in their own history as seen in [Figure 37](#) and [Figure 40](#). **Telecoms, Consumer Goods and Healthcare** stand out as the main Non-Cyclical risks to watch, while **consumer-oriented Cyclical sectors and Transports** are also displaying higher leverage.

Figure 40: While Transports, Media, and Telcos represent the largest near term risks, a sustained energy shock could impact Consumer-oriented Cyclicals more



Source : Deutsche Bank, S&P, Bloomberg Finance LP, ICE indices

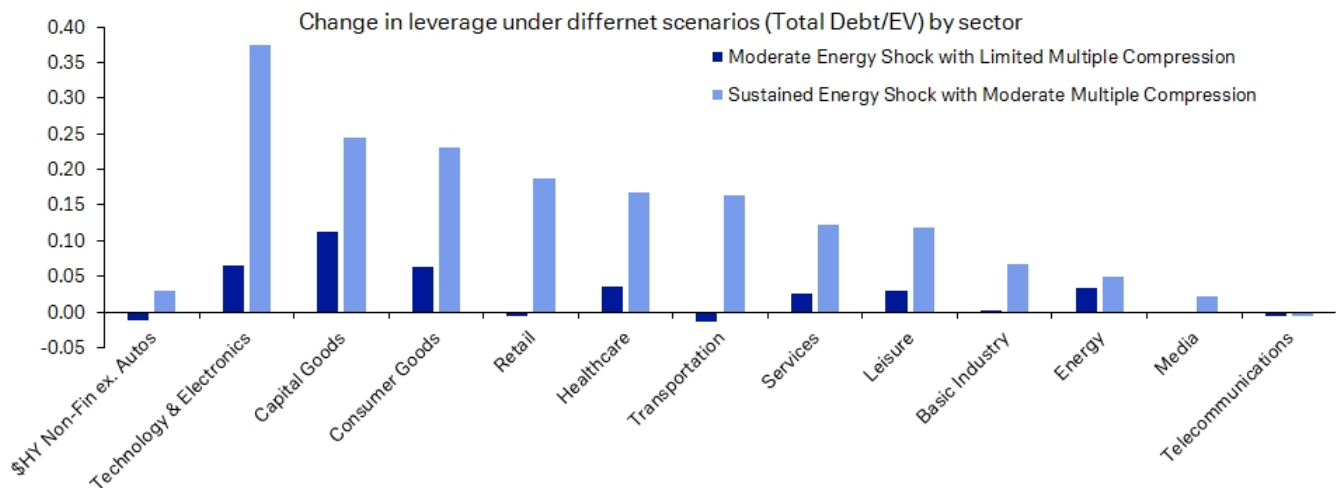
Moderate Energy Shock with Limited Multiple Compression = Margins compress to Min(current, 75th %ile since 2004) and Multiple compress to Min(current, 75th %ile since 2004).
Sustained Energy Shock with Moderate Multiple Compression = Margins compress to Min(current, 33rd %ile since 2004) and Multiple compress to Min(current, Median since 2004)



Under our first scenario, the moderate commodity shock with limited multiple compression, **lowly-levered cyclicals see the biggest jump in leverage** on the back of both compressed margins and re-rated multiples. \$HY Capital Goods, Technology, and Consumer Goods see the biggest jump in leverage.

Capital Goods and Technology currently enjoy relatively low leverage levels, but come under pressure quickly even in the more mild scenario. While there is some dispersion under the first scenario, the second more sustained price shock leads to a wider spread increase in Spec-Grade leverage across sectors. Nonetheless, the more adverse scenario still sees Cyclical and Consumer-Oriented sectors underperform.

Figure 41: Technology, Cap Goods, and Consumer Goods would see the biggest impact from both a moderate and sustained energy shock and higher-for-longer rates compressing multiples

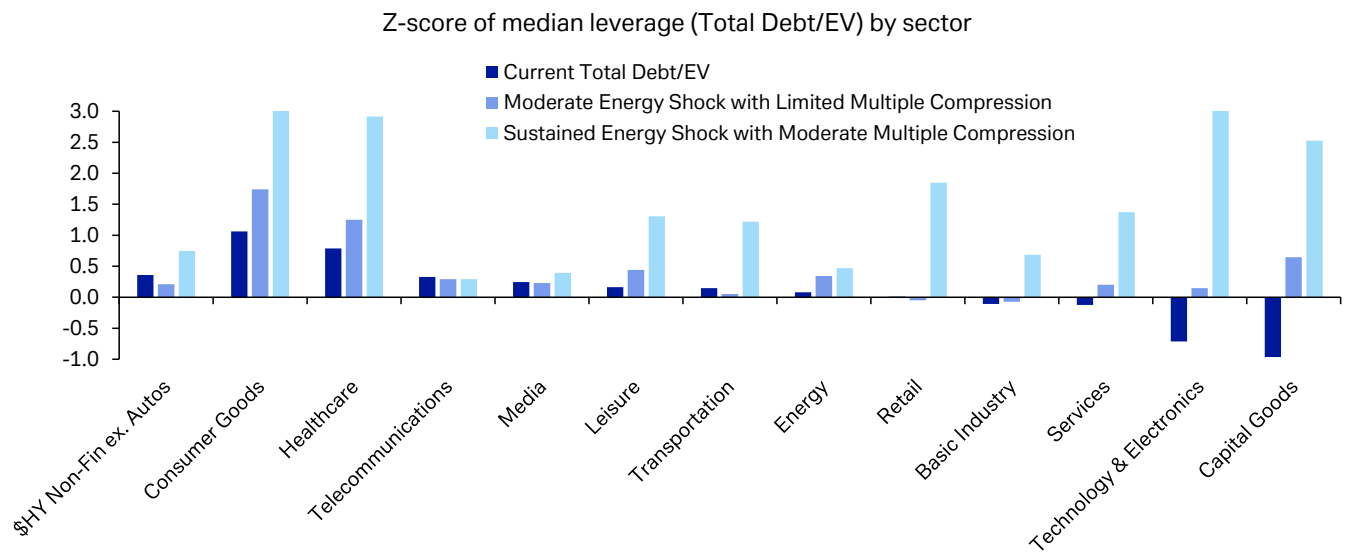


Source : Deutsche Bank, S&P, Bloomberg Finance LP, ICE indices
Moderate Energy Shock with Limited Multiple Compression = Margins compress to Min(current, 75th %ile since 2004) and Multiple compress to Min(current, 75th %ile since 2004).
Sustained Energy Shock with Moderate Multiple Compression = Margins compress to Min(current, 33rd %ile since 2004) and Multiple compress to Min(current, Median since 2004)

Viewing the scenario analysis in Z-score space offers a potential view into how risks may appear in the \$LL and private-credit markets. Since we do not have visibility into how enterprise values are developing in these markets, tracking the trends in public HY markets can inform us on where risks are trending. Using the framework and scenario analysis above, **Non-Cyclicals are the main risks to monitor in the current environment**, with greatest risk in Consumer Goods, Healthcare, and Telecoms. **US Cyclicals would face much greater risks in adverse scenarios** if multiples contract and rising energy costs cause margins to compress substantially, though Healthcare and Consumer goods would still be large risks.



Figure 42: When looking at leverage in their own histories, Non-Cyclical sectors such as Consumer Goods, Healthcare, and Telcos are high, with the former two sectors seeing more stress under our adverse scenarios as well



Source : Deutsche Bank, S&P, Bloomberg Finance LP, ICE indices
Moderate Energy Shock with Limited Multiple Compression = Margins compress to Min(current, 75th %ile since 2004) and Multiple compress to Min(current, 75th %ile since 2004) .
Sustained Energy Shock with Moderate Multiple Compression = Margins compress to Min(current, 33rd %ile since 2004) and Multiple compress to Min(current, Median since 2004)



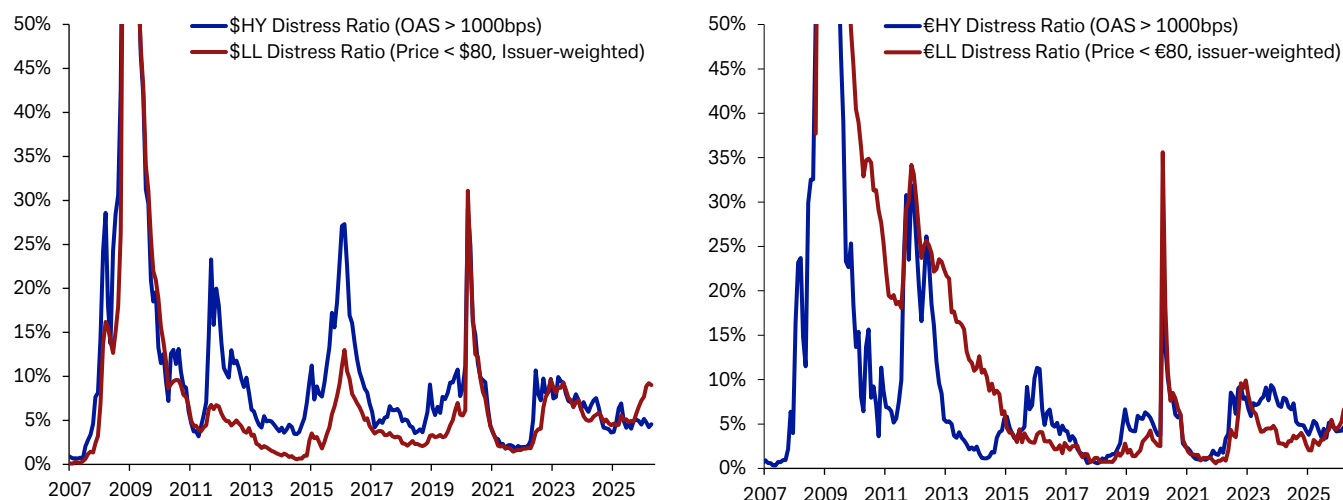
What's priced into credit?

LL distressed ratios are rising while HY ratios have moderated

How do our forecasts for US and EUR spec-grade default rates over the next 12 months compare to what investors are pricing into Leveraged Finance markets currently? We anticipate that US and European defaults will slow to 3.6% by year-end before climbing slowly back to 4% in the US and 3.9% in Europe by this time next year. **While overall index spreads are trading near historically tight levels, distress ratios are not similarly near their lows.** This reflects that investors have deemed pockets of debt quite risky and historically elevated distress ratios have been a good indicator of restructuring activity in 9-10 months.

\$HY and \$LL distress ratios started to diverge in October of last year. This coincides with investors becoming more wary of private credit risks following the events surrounding Tricolor and First Brands. The level of distressed issuers rose as AI-disruption became a theme early this year, sending some \$LL Software prices significantly lower. Through all of this, \$HY distress ratios have remained resilient near post-GFC lows, when discounting the 2021-22 period when there was ample fiscal & monetary support. **This reflects the nature of where defaults are likely to be concentrated in the coming months**, as LL markets currently have more of the structural risks we outlined above (more software and lower rated). €LL markets are smaller than the other spec-grade markets, but there is a similar dynamic of increasing floating-rate risk relative to €HY markets where distress ratios are pricing in less imminent default risk. Nonetheless, €Spec-Grade distress ratios are elevated relative to the post-European Debt Crisis lows, as investors expect moderate defaults moving forward.

Figure 43: HY index distressed ratios have decreased over the last few years as index quality improved and issuer stress became more localized; meanwhile LL distress ratios have trended higher in recent months due in part to percolating software and private credit worries



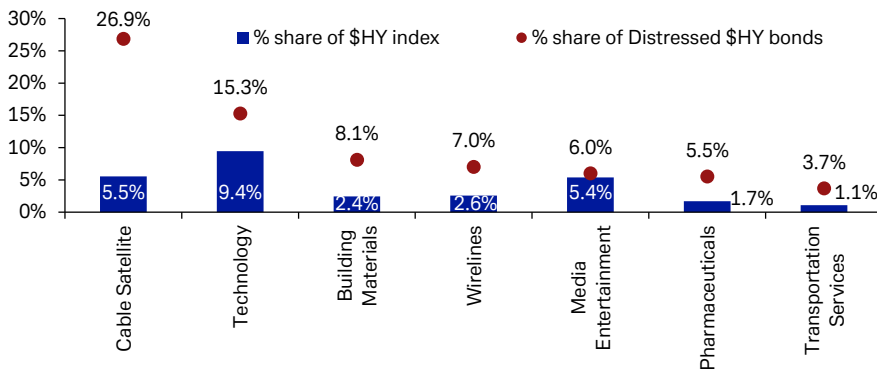
Source : Deutsche Bank, S&P, Bloomberg Finance LP, ICE Indices



USD Distressed Debt is highly concentrated, while there is greater dispersion in EUR markets

Delving into where distressed debt resides in each market, **risks are far more concentrated in USD Spec-Grade markets**. Technology/Software has the largest concentration of distressed debt, with significant Telecom risks a close second. Further away, there are signs that Consumer Cyclical as well as Healthcare risks are percolating. Within \$HY bonds, Cable & Satellite (27%) makes up over a quarter of distressed debt. Adding in Wirelines (7%), means that Telecoms make up over a third of \$HY debt trading wider than 1000bps. The recent software weakness weighs on \$HY Technology (15%) which has the second largest sum of distressed bonds with Media (6%) the fifth largest but nearly in-line with the sector's overall weight in the index.

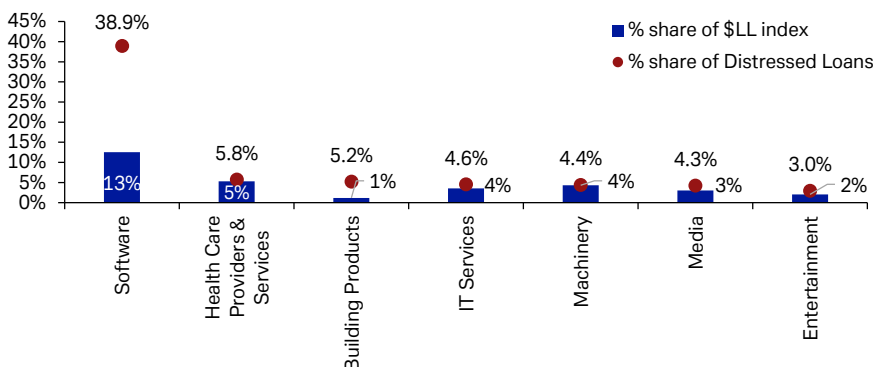
Figure 44: \$HY Cable Satellite is the main sector with an outsized share of distressed debt followed by Technology.



Source : Deutsche Bank, Bloomberg Finance L.P.

In \$Leveraged Loans, the Software sector remains the main story. It makes up roughly 13% of the \$LL market and has nearly two-fifths of the distressed debt. At nearly 39%, its share of distressed loans is basically triple its weight in the index. Building Materials is the only other \$LL sector with outsized risk priced in, but it is a much smaller sub-sector.

Figure 45: Software continues to have the largest share of distressed debt in the \$Lev Loan market.



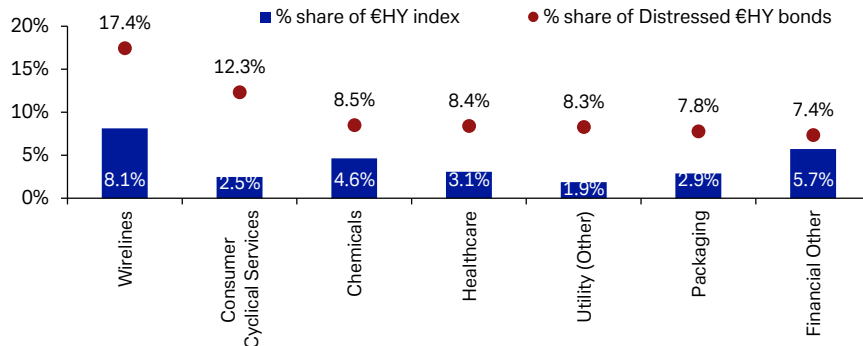
Source : Deutsche Bank, Pitchbook

Within €HY & €LL, the risks are far more dispersed. Risks are also being priced into



both Cyclical and Non-Cyclicals. Wireless issuers have over 17% of distressed bonds to lead €HY, with fellow Non-Cyclical sectors Healthcare (8%) and Non-Electric Utilities (8.3%) also in the top 5. On the Cyclical side, Consumer Cyc Services (12%), Chemicals (9%), and Packaging (8%) round out the other €HY subsectors with significant outsized priced-in default risk.

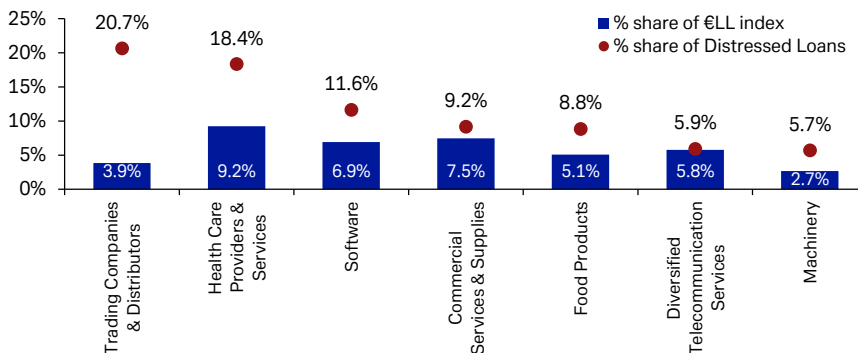
Figure 46: €HY distressed bonds are spread between Cyclical and Non-Cyclical issuers, with the greatest share in Wirelines



Source : Deutsche Bank, Bloomberg Finance L.P.

€Leveraged Loan distress debt is a much smaller market and the risks are more concentrated. Trading Companies and Distributors (21%), which highlights how soaring energy prices and supply chain issues could be impacted by the conflict in the Middle East. Healthcare (18%) has a slightly lesser share of distressed loans as Trading Companies, with Software (11.6%) the next largest concentration.

Figure 47: €LL distressed loans are more concentrated in Trading Companies and Healthcare, with some Software exposure as well



Source : Deutsche Bank, Pitchbook



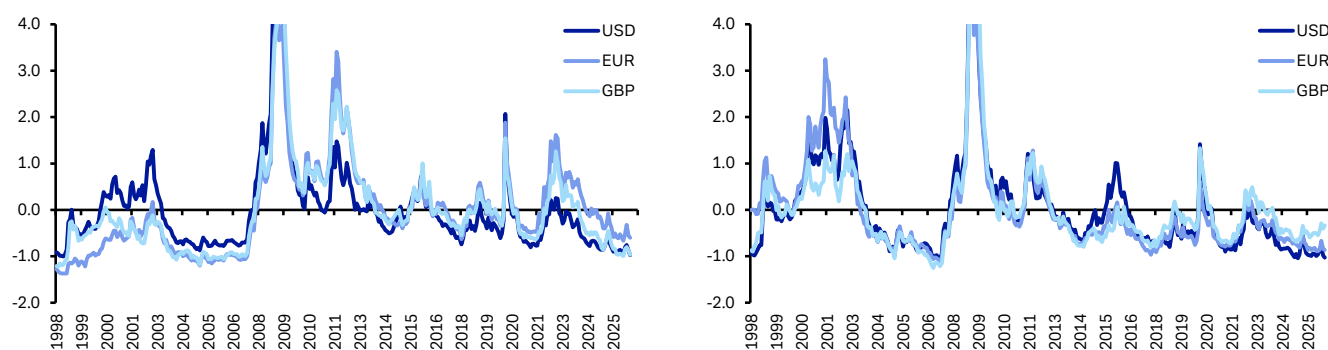
Non-financial (cash) spread implied default rates

This final section was the initial focus of these annual studies. The purpose was to assess the level of credit spreads required to compensate buy-and-hold HY investors for default risk. We also analyze what level of defaults are currently implied by spreads, and how much excess spread is being provided to investors under different default scenarios.

The analysis focuses on Non-Financials around the 5-year tenor. To achieve this, we include all senior bonds 4-6 years until maturity within the ICE indices. Having calculated the average spreads for our sample bond sets, we use three different recovery assumptions (0%, 20% and 40%) to calculate implied default rates. We then compare these implied default rates with the appropriate average five-year cumulative global default rate, as well as the worst five-year periods observed since 1981. This is also compared against the average and worst five-year level for the primary region of the various bond markets (Europe – EUR; UK – GBP; US – USD). We have highlighted in [Figure 49](#) the cells where the implied default rate is lower than the average experience (darker blue) as well as those where the implied default rate is lower than the worst but higher than the average (lighter blue). The shading is based on regional default rates. This analysis should be viewed from the perspective of a longer-term buy-and-hold investor.

HY spreads across USD and EUR in particular are trading near multi-decade tights. The Q1 volatility stemming from AI-disruption fears and then the fallout of the US-Iran war has dissipated, with **spreads returning to near their long-term tights**. The solid economic growth that we referenced above as well as elevated yields have meant investors are looking through some of the percolating risk factors. Given the starting point for spreads, we do expect greater decompression and dispersion as margin pressures and higher-for-longer rates set in.

Figure 48: Z-scores of IG (left) and HY (right) credit spreads since 1998 highlight just how tight valuations currently are, especially USD and EUR HY markets



Source : Deutsche Bank, Bloomberg Finance LP, ICE Indices

With the current rich valuations in mind, we look at how spread implied defaults compare to past periods. As to be expected, IG-rated implied default rates are going to be well above even the worst observed defaults for all recovery assumptions. Current \$IG spreads imply a 3.9% cumulative five-year default rate, assuming a 40% recovery, and a 6.4% default rate with 0% recovery. This premium is due to spreads also pricing in some liquidity and downgrade risks that are not



captured by the implied default equation we lay out in the appendix.

Even as overall HY index spreads are near historical tights, current 5yr spreads are still pricing in an above average default rate across markets. This is even when we assume 40% recoveries. **Assuming higher recoveries, the implied 5yr cumulative HY default rate is around 20% or better across all three markets, which is still above the average cumulative five-year default rate in each region by a decent margin.** Though it remains well below the worst-case scenario defaults, which would approximate a recessionary environment in the next five years. Current **CCC spreads in both USD and EUR are too tight** to adequately compensate investors for average historical cumulative 5 year defaults assuming a 40% recovery rate. If recovery rates fall to near 20%, then not only do current CCC spreads not compensate investors but \$HY and \$B spreads are too tight to protect investors.

Figure 49: Non-Financial 5-year cumulative spread implied default rates based on different recovery assumptions

		Spread Implied 5yr Cumulative Default Rate				Actual 5yr Cumulative Default Rates			
		5yr Spread	0% Recovery	20% Recovery	40% Recovery	Global Worst	Global Average	Regional Worst	Regional Average
EUR	IG	78	3.9%	4.8%	6.4%	2.9%	0.8%	2.4%	0.3%
	AA	44	2.2%	2.8%	3.7%	0.5%	0.1%	0.0%	0.0%
	A	64	3.2%	4.0%	5.3%	1.2%	0.3%	1.8%	0.2%
	BBB	90	4.4%	5.5%	7.3%	5.0%	1.6%	5.9%	1.0%
	HY	295	13.5%	16.6%	21.5%	32.5%	16.5%	38.8%	14.7%
	BB	214	10.0%	12.3%	16.0%	20.0%	7.9%	27.3%	7.3%
	B	382	17.2%	21.0%	27.0%	39.9%	19.3%	47.6%	17.5%
CCC		753	29.5%	35.5%	44.2%	68.5%	50.5%	64.0%	41.5%
USD	IG	64	3.1%	3.8%	5.1%	2.9%	0.8%	2.4%	0.9%
	AA	29	1.4%	1.8%	2.3%	0.5%	0.1%	0.6%	0.1%
	A	46	2.2%	2.7%	3.6%	1.2%	0.3%	1.3%	0.3%
	BBB	79	3.8%	4.8%	6.3%	5.0%	1.6%	4.3%	1.6%
	HY	268	12.3%	15.2%	19.7%	32.5%	16.5%	30.9%	16.7%
	BB	167	7.9%	9.8%	12.8%	20.0%	7.9%	17.6%	7.8%
	B	327	14.9%	18.3%	23.6%	39.9%	19.3%	38.3%	19.0%
CCC		746	30.1%	36.1%	44.9%	68.5%	50.5%	72.5%	52.3%
GBP	IG	79	3.9%	4.8%	6.4%	2.9%	0.8%	3.6%	0.4%
	AA	16	0.8%	1.0%	1.3%	0.5%	0.1%	0.0%	0.0%
	A	61	3.0%	3.8%	5.0%	1.2%	0.3%	3.2%	0.3%
	BBB	94	4.6%	5.7%	7.5%	5.0%	1.6%	7.3%	1.0%
	HY	402	17.8%	21.8%	27.9%	32.5%	16.5%	42.9%	16.1%
	BB	329	14.6%	17.9%	23.1%	20.0%	7.9%	30.0%	5.5%
	B	444	19.8%	24.1%	30.8%	39.9%	19.3%	42.9%	15.6%

Source : Deutsche Bank, ICE Indices, S&P

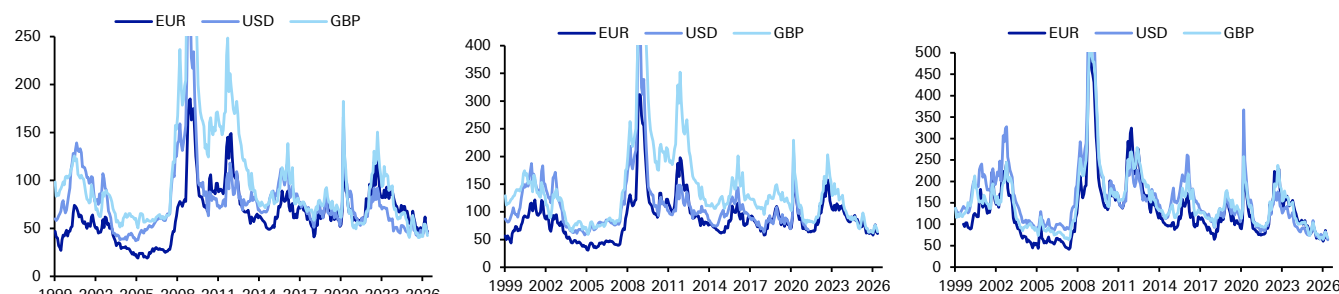
Note: Darker shading means implied default rate is lower than long-term global average. Lighter shading means implied default rate is lower than the worst historical global rate. As of 5/28/26



Non-financial (cash) default spread premiums (DSP)

Default spread premiums (DSP) provide us with a sense of the excess spread offered in credits over that which is required to compensate for different default scenarios. We start by looking at DSP histories by rating in the EUR, USD and GBP credit markets. They are calculated by subtracting the spread required to compensate for average (historical) default rates and assuming an average recovery rates (40%) from the index spread level. In [Figure 50](#) we show this for IG ratings, where DSPs have always been positive.

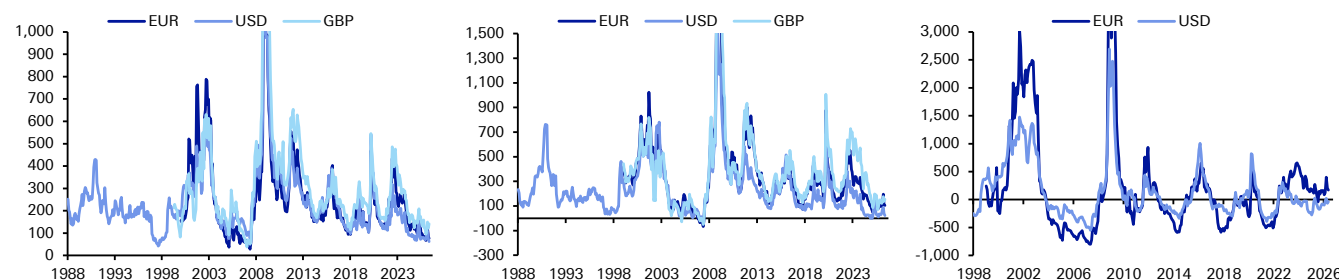
Figure 50: IG DSPs By currency and rating (bps) – AA (left), A (middle) and BBB (right)



Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE indices

In [Figure 51](#), we show the same for ratings across HY. With spreads near historic tights, it follows that default spread premiums are near the lows of their long-term ranges. This indicates that **investors are being compensated for very little outside of default risk**. We appear to be in a period where liquidity, rating transition, and recovery uncertainty premia are being heavily discounted. The only ratings where DSPs are not at historic lows are EUR and USD CCCs. The former is a relatively small and idiosyncratic market, while the latter's DSP is still negative, meaning **\$CCC spreads are not compensating for average default risk and average recoveries** at current levels for buy and hold investors.

Figure 51: HY DSPs by currency and rating (bps) – BB (left), B (middle) and CCC (right)



Source : Deutsche Bank, Markit Group, ICE indices

Focusing on the current market pricing, we consider **which rating buckets are offering the greatest DSP given different default and recovery scenarios**. Once again, this analysis is based around the five-year maturity point (focused on bonds with 4-6 years to maturity) and not using index spreads as a whole as we do in Figures 61 and 62. In addition to average default cycles, we also consider the worst historical cycle for each market as well as the most benign. Additionally, we introduce two more recovery assumptions (0% and 20% in addition to 40%). This framework essentially tries to assess the most attractive default-adjusted spread

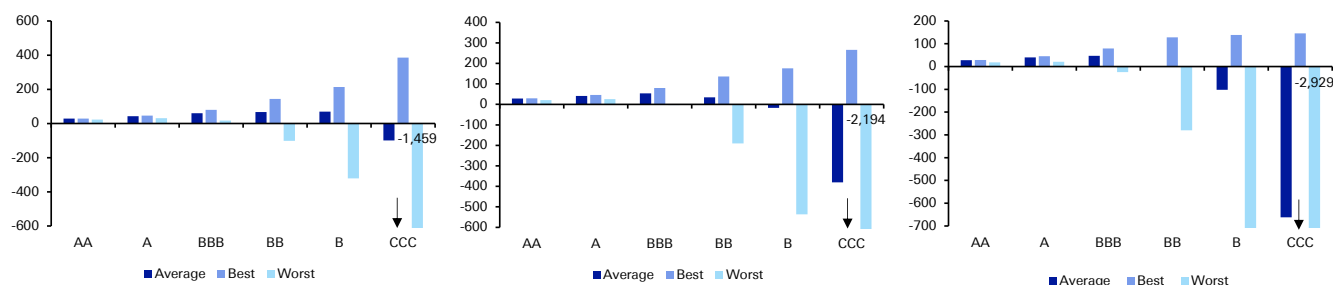


from a buy-and-hold perspective.

We start by looking at USD credit in [Figure 52](#) and provide some of the key highlights below.

- Based on an average default cycle and 40% recovery, **single-Bs offer the widest DSP**, with BBBs seeing the largest premium if recovery levels were to fall.
- Under the worst-case default scenario, which would be a recessionary default cycle, IG generally offers the widest DSP under all recovery assumptions, with even BBs negative at 40% recovery.
- **CCCs only outperform from here under the most bullish scenarios**, wherein they offer a sizeable premium. But of course in the worst-case scenario, there is a substantial wipeout of returns. As laid out above, **current CCC DSPs are negative for an average historical default rate** even at a 40% recovery assumption.

Figure 52: USD non-financial default spread premiums for average, best (lowest) and worst (highest) 5-year cumulative default observations assuming 40% (left), 20% (middle) and 0% (right) recoveries

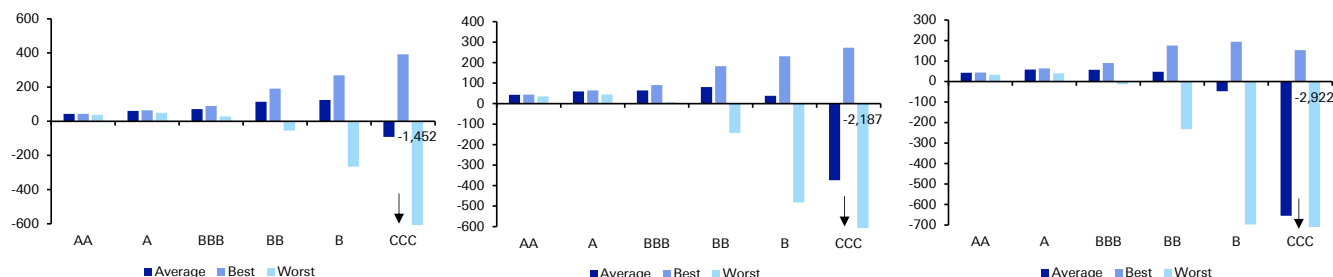


Source : Deutsche Bank, ICE indices

The EUR market analysis, laid out in [Figure 53](#), has broadly similar conclusions.

- **Single-Bs provide the widest DSP** based on an average default cycle for the 40% recovery assumption scenario, single-As actually provide the largest premium if recoveries are 0%, while BBs offer the best at 20%. This speaks, in part, to how tight EUR BBBs trade.
- **CCC DSPs are negative even if recoveries are 40%** under average default scenarios, and grows substantially if recoveries are lower than that.

Figure 53: EUR non-financial default spread premiums for average, best (lowest) and worst (highest) 5-year cumulative default observations assuming 40% (left), 20% (middle) and 0% (right) recoveries



Source : Deutsche Bank, ICE indices

While we believe that defaults can moderate in the short-run, there are still risks we



will see elevated cumulative defaults going forward. Especially if the Fed is not going to be easing policy in the short-to-medium term and there is potentially less fiscal support ahead if Congress becomes split in the latter half of President Trump's second term. Higher-quality, high yield bonds can offer a good level of buy-and-hold value despite potential spread volatility.



Cohort analysis

In this sub-section, we update our analysis assessing what's priced in to current spreads by comparing them with the spreads that would have been required to compensate for default based on all realized 5-year cohorts over the last 30+ years using different recovery assumptions. We look at this on a global basis.

Global cohorts

We start by looking at defaults globally with the analysis based only on non-financial issuers. We calculate spreads required to compensate for default based on each actual realised 5-year discreet cohort formed since 1981 (based on S&P data). Overall, this analysis is useful for showing where current spreads are across the board and in which periods through history these spreads would have (and would not have) compensated for the risk of default and by how much. We use both 40% and 0% recoveries, as exact recoveries by cohort are not available. This provides us with a compensation range if readers want to estimate for a different recovery rate.

In [Figure 55](#), we show the spread required to compensate for each historical cohort across both IG and HY rating bands. We have drawn a box around each cell where the spread required to compensate for default is wider than the tightest current spreads (as shown in [Figure 54](#)) of the relevant rating band across currencies (USD, EUR, GBP).

Figure 54: Current Non-financial spreads (vs. governments) by rating (bps)

	Currency	IG	AA	A	BBB	HY	BB	B	CCC
Current	EUR	75	48	65	85	269	171	377	1,338
	USD	72	47	57	89	271	161	295	936
	GBP	81	44	70	94	474	236	426	
Wide since 2007	EUR	380	195	321	548	1,927	1,401	2,459	6,160
	USD	524	320	460	650	1,660	1,234	1,825	2,992
	GBP	360	205	315	542	2,585	2,167	3,520	
Tight since 2007	EUR	59	24	52	70	212	142	260	475
	USD	72	41	56	89	258	156	239	395
	GBP	76	31	66	88	173	127	215	

Source : Deutsche Bank, Bloomberg Finance LP, ICE Indices
Data as of 5/28/2026

When looking at global defaults, the IG analysis as always shows that current spread levels across all ratings are wider than the spread needed to compensate for each of the past 5-year default cycles. As discussed, this is because of the other risks priced into spreads. Though it speaks to how tight BBBs are trading, that at a 0% recovery assumption, current BBBs would not compensate default risks around the dot-com era.

The HY market provides far more interesting observations for investors. Of the 40 cohorts that we have data for, going back to 1982, current CCC spreads would have adequately compensated investors for nearly half (21) of the previous cohorts when assuming a 40% recovery. **For HY overall, current spreads would not have compensated for 11 previous cohorts**, which are grouped around '85-'90 and the dot-com bubble burst.

For a more extreme zero recovery assumption and with spreads near long-term tightness, current valuations would only compensate investors for cohorts which saw historically few defaults. This would be the periods like the mid-90s, post-dot-com era, post-GFC, and the recent post-Covid cohort. These are also cohorts which benefitted from easier monetary policy. Only in 6 of 40 cohorts would CCCs have



been adequately compensated for the ensuing default risks at current spreads if recoveries were zero.

While we are not forecasting a dire default cycle ahead, there are concrete risks to keep cumulative defaults materially above the best-case periods of the past where default rates bottomed out. Therefore this exercise indicates that **current CCC spreads are too optimistic**. B3s are also likely priced for perfection while higher-rated spec-grade debt can outperform if there is no recession on the horizon. While our modelled 3-year cumulative default rates are likely to stay higher than average, the lack of recessionary risks mean current cohorts are not likely to reach the extremes seen of the 90s or dot-com era.

Figure 55: Spreads required to compensate for default based on Global 5-year cumulative default rates by cohort (boxed numbers show past 5-year default periods where current spreads would not compensate for eventual default risk)

	40% Recovery								0% Recovery							
	IG	AA	A	BBB	HY	BB	B	CCC	IG	AA	A	BBB	HY	BB	B	CCC
1981	8	0	3	25	166	166	192		13	0	5	41	277	277	319	
1982	11	0	8	27	235	224	228	471	19	0	14	46	391	374	380	784
1983	12	6	3	32	214	181	213	655	19	11	5	53	356	301	355	1,091
1984	9	5	3	23	269	188	320	750	15	8	5	39	448	313	533	1,250
1985	5	5	6	5	275	172	379	360	8	8	9	8	458	287	632	600
1986	13	5	8	33	241	98	342	631	22	8	13	55	401	164	570	1,052
1987	22	5	15	52	275	141	289	921	37	8	25	86	459	235	482	1,535
1988	15	5	10	34	307	152	353	923	25	8	16	57	512	253	588	1,538
1989	10	0	0	36	353	186	403	1,108	17	0	0	60	589	311	672	1,846
1990	8	0	0	26	353	148	434	1,630	13	0	0	44	589	246	723	2,717
1991	5	0	3	13	276	70	365	1,031	9	0	5	22	460	116	608	1,719
1992	3	0	3	4	148	24	207	657	4	0	5	7	247	41	345	1,095
1993	2	0	3	3	132	49	196	462	4	0	5	6	221	82	326	769
1994	3	0	3	7	136	58	196	772	6	0	5	11	227	96	326	1,286
1995	8	0	0	21	144	86	173	831	13	0	0	35	240	143	288	1,385
1996	8	0	3	18	169	91	242	404	14	0	4	30	282	152	404	674
1997	15	0	7	29	265	144	369	1,292	25	0	12	48	442	240	614	2,154
1998	31	0	13	58	395	218	543	1,698	52	0	22	97	659	363	905	2,830
1999	32	0	12	61	491	268	648	1,891	54	0	21	102	819	446	1,079	3,152
2000	33	0	15	58	476	244	618	2,030	55	0	25	97	793	407	1,030	3,383
2001	35	0	10	62	438	205	557	2,205	59	0	17	104	730	342	928	3,675
2002	19	0	0	36	308	111	352	1,542	31	0	0	60	513	185	587	2,570
2003	4	0	0	8	161	48	162	1,000	7	0	0	13	268	80	270	1,667
2004	6	0	2	9	105	44	113	554	9	0	3	15	175	73	188	923
2005	13	0	2	23	157	85	191	556	22	0	3	38	262	141	319	927
2006	8	0	2	13	188	83	239	731	13	0	3	21	314	138	398	1,218
2007	6	0	0	11	218	93	279	908	11	0	0	19	363	154	465	1,513
2008	5	0	0	9	267	85	351	1,677	9	0	0	15	445	141	585	2,795
2009	2	0	0	3	263	49	298	2,017	3	0	0	4	438	81	497	3,361
2010	0	0	0	0	125	23	114	649	0	0	0	0	208	38	191	1,081
2011	1	0	2	0	121	38	133	791	1	0	4	0	202	63	222	1,319
2012	4	0	2	5	142	61	145	1,015	6	0	4	8	237	101	242	1,692
2013	2	0	0	3	156	46	177	933	4	0	0	6	261	77	295	1,555
2014	4	0	0	7	153	46	180	769	7	0	0	11	255	77	301	1,282
2015	4	0	0	6	166	47	195	922	7	0	0	11	277	79	325	1,536
2016	8	0	0	13	190	36	219	1,200	14	0	0	21	317	60	365	2,000
2017	5	0	0	8	161	27	171	989	8	0	0	13	268	45	284	1,648
2018	5	0	4	5	165	30	174	1,317	8	0	7	9	274	49	290	2,195
2019	4	0	0	6	180	36	196	1,292	7	0	0	10	300	59	327	2,153
2020	5	0	0	7	185	45	163	1,780	8	0	0	12	308	76	272	2,967
2021	3	0	0	4	132	29	109	513	5	0	0	7	221	49	181	855

Source: Deutsche Bank, S&P, Bloomberg Finance L.P., ICE Indices



Appendix A

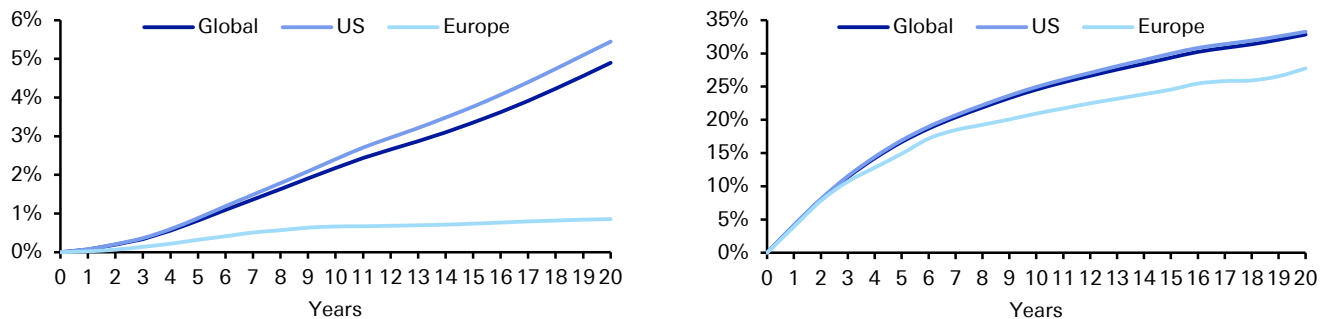
Methodology, raw data and calculations for HTM analysis

The spread required to compensate for default is calculated using the expected loss for each rating band given historical default and different assumed recoveries. The starting point for such calculations is the rating agencies' historical databases on default. For the purposes of this report, we have mainly used data from S&P, as it provides us the flexibility to focus purely on non-financial defaults as well as looking at defaults regionally, although much of our analysis focuses on the global numbers.

US vs. Europe cumulative default rates

In [Figure 56 - Figure 58](#) we show a number of charts comparing cumulative default rates for European issuers with those of US issuers. The main point here is that, in general, average historical European default rates have been lower. This is in large part due to the fact there is much less history for rated European issuers and less CCC rated issuers specifically. Based on the S&P database, there has only been more than 100 European IG rated issuers since 1995, and for HY there has only been more than 100 since 2004. Therefore, most of the relevant data for Europe occurs during the past decade or so of extremely low defaults, which probably helps explain the lower default rates relative to the more developed US market. This is also why we have focused on the global numbers for much of our analysis. Over time we would expect European markets to converge further to US markets.

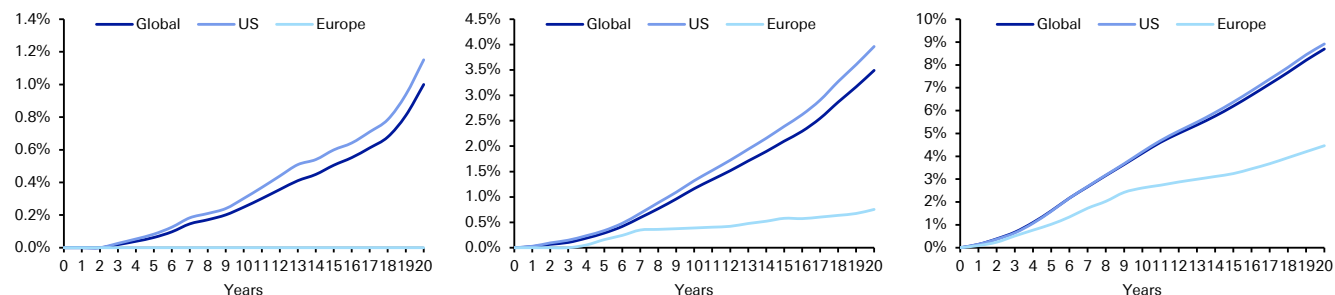
Figure 56: S&P Average Cumulative IG (left) and HY (right) Default Rates



Source : Deutsche Bank, S&P

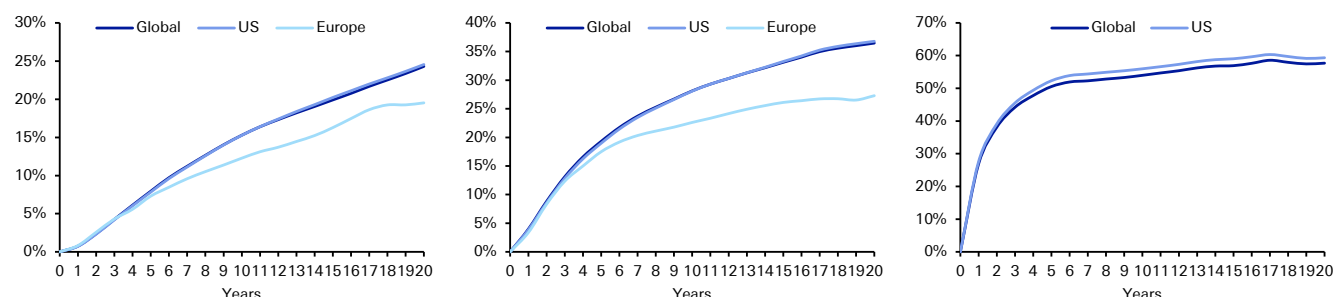


Figure 57: S&P Average Cumulative AA (left) A (middle) and BBB (right) Default Rates



Source : Deutsche Bank, S&P

Figure 58: S&P Average Cumulative BB (left) B (middle) and CCC (right) Default Rates



Source : Deutsche Bank, S&P

Calculating spreads required to compensate for default

The spreads calculated are spreads based on the maturity of the indices. That would imply we need to calculate the spreads required to compensate for cumulative default probabilities over that time period. The formula we use assumes constant instantaneous hazard rate, which is determined by the spread to that maturity. We then integrate the continuously compounded survival probability to get the cumulative survival probability over time. Thus, the formula can be used to calculate the spread from cumulative default probability or vice versa. We get the cumulative default probabilities from the S&P database and calculate the compensation spreads from them.

$$D = 1 - e^{-ST/(1-R)}$$

where D is the cumulative default probability over time T and S is the Spread to that maturity and R is the recovery

Solving for spread S,

$$S = -\frac{(1-R)}{T} \ln(1-D)$$

Some of our analysis uses different calculations where it makes more sense to look at the path of defaults, rather than simply assuming a constant instantaneous hazard rate.

Recovery rates

In terms of recovery, we have used a selection of different rates throughout the note, generally focusing on 0%, 20% and 40%. However, in [Figure 59](#) (for comparison



purposes) we publish a table showing average senior unsecured recovery rates for different rating bands. Recovery rates can fluctuate depending on different macroeconomic backdrops, which is why we include the range of recovery scenarios. Recoveries are a very important factor when trying to assess losses in the case of default and therefore calculate the spread required to compensate for default.

Figure 59: Moody's Senior Unsecured Recovery Rates by Rating (1983-2025)

	Year 1	Year 2	Year 3	Year 4	Year 5	Average
Aa	37.2%	39.0%	38.1%	44.0%	43.2%	40.3%
A	37.7%	46.2%	47.4%	46.8%	46.2%	44.9%
Baa	38.4%	40.3%	41.0%	41.3%	41.4%	40.5%
Ba	40.1%	40.3%	40.0%	40.0%	39.9%	40.1%
B	36.5%	36.4%	37.2%	37.9%	38.5%	37.3%
Caa-C	38.9%	39.0%	38.8%	38.8%	38.7%	38.8%
IG	38.2%	41.6%	42.7%	43.1%	43.0%	41.7%
HY	38.3%	38.2%	38.3%	38.5%	38.7%	38.4%
All	38.3%	38.4%	38.5%	38.8%	39.1%	38.6%

Source : Moody's

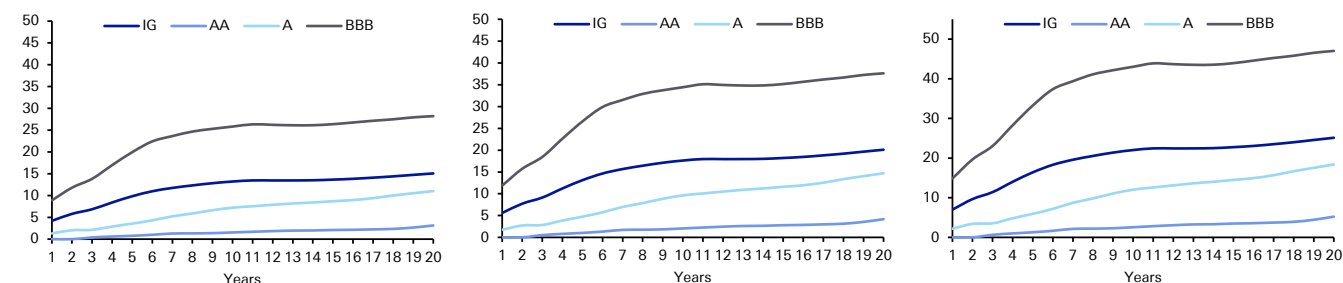
Spreads required to compensate for default

Figure 60: Spreads Required to Compensate for Default Assuming Different Recovery Levels

Years	40% Recovery									20% Recovery									0% Recovery								
	IG	AA	A	BBB	HY	BB	B	CCC	IG	AA	A	BBB	HY	BB	B	CCC	IG	AA	A	BBB	HY	BB	B	CCC			
1	4	0	1	9	252	46	247	1,701	6	0	2	12	336	62	329	2,267	7	0	2	15	420	77	412	2,834			
2	6	0	2	12	250	73	280	1,283	8	0	3	16	333	98	373	1,711	10	0	3	20	416	122	466	2,139			
3	7	0	2	14	241	88	284	1,035	9	1	3	18	322	118	379	1,380	11	1	4	23	402	147	473	1,725			
4	8	1	3	17	230	97	275	875	11	1	4	23	307	129	366	1,167	14	1	5	28	384	161	458	1,458			
5	10	1	4	20	219	101	261	772	13	1	5	27	292	135	348	1,030	16	1	6	33	365	169	436	1,287			
6	11	1	4	22	207	104	248	668	15	1	6	30	276	139	330	890	18	2	7	37	345	173	413	1,113			
7	12	1	5	24	195	104	234	590	16	2	7	32	260	139	312	787	20	2	9	39	325	173	389	983			
8	12	1	6	25	185	104	220	525	16	2	8	33	247	138	294	700	21	2	10	41	308	173	367	875			
9	13	1	7	25	177	103	209	487	17	2	9	34	236	137	279	649	21	2	11	42	295	172	349	812			
10	13	2	7	26	169	102	200	450	18	2	10	34	225	136	266	601	22	3	12	43	282	170	333	751			
11	13	2	8	26	162	99	191	418	18	2	10	35	216	132	254	558	22	3	13	44	269	166	318	697			
12	13	2	8	26	155	97	182	394	18	2	10	35	206	129	243	525	22	3	13	44	258	161	304	656			
13	13	2	8	26	149	95	175	373	18	3	11	35	199	126	234	497	22	3	14	44	248	158	292	621			
14	14	2	8	26	144	92	169	353	18	3	11	35	191	123	225	470	23	3	14	44	239	154	281	588			
15	14	2	9	26	139	91	163	332	18	3	12	35	185	121	218	442	23	3	14	44	232	151	272	553			
16	14	2	9	27	135	89	159	317	18	3	12	36	180	119	212	422	23	4	15	45	225	149	265	528			
17	14	2	9	27	130	88	153	295	19	3	13	36	173	117	204	393	24	4	16	45	217	147	255	491			
18	14	2	10	27	126	86	147	276	19	3	13	37	167	115	196	368	24	4	17	46	209	144	245	460			
19	15	3	11	28	122	86	142	262	20	4	14	37	163	114	189	349	25	4	18	47	204	143	237	437			
20	15	3	11	28	119	85	138	251	20	4	15	38	159	114	183	335	25	5	18	47	199	142	229	419			

Source : Deutsche Bank, S&P

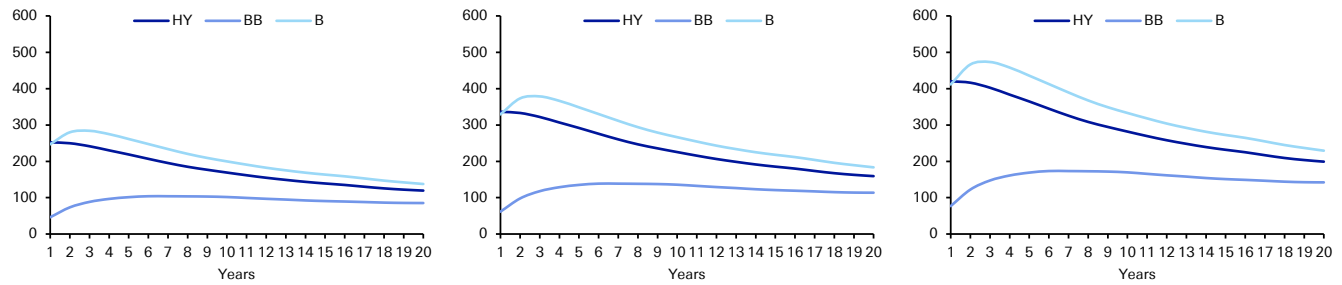
Figure 61: IG Spread Required to Compensate for Default Assuming a 40% (left), 20% (middle), 0% (right) Recovery



Source : Deutsche Bank, S&P



Figure 62: HY Spread Required to Compensate for Default Assuming a 40% (left), 20% (middle), 0% (right) Recovery



Source : Deutsche Bank, S&P

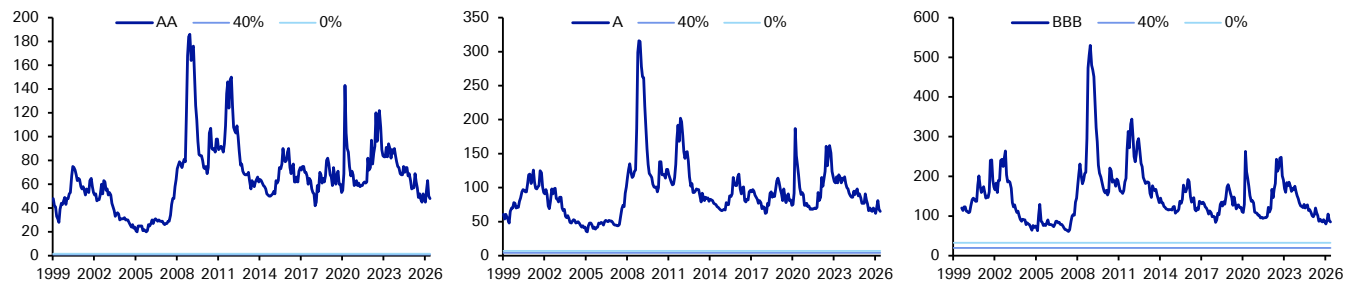
Credit spreads vs. spreads required to compensate for default

Whilst we have formed most of our conclusions in earlier sections of this note, the following section takes a more detailed graphical look at the spreads required to compensate for default probability assuming different recovery rates against historical index level spreads.

The charts focus on non-financial indices across the rating bands. We include charts for the EUR, USD, and GBP markets across both IG and HY. We initially show the spread histories compared with the spread required to compensate for average historical default based on the appropriate rating and index average life, assuming both a 40% and 0% recovery. We then show the DSP charts where we subtract the spread required to compensate for default from the actual trading levels. For this we only use the analysis where we assume a 40% recovery.

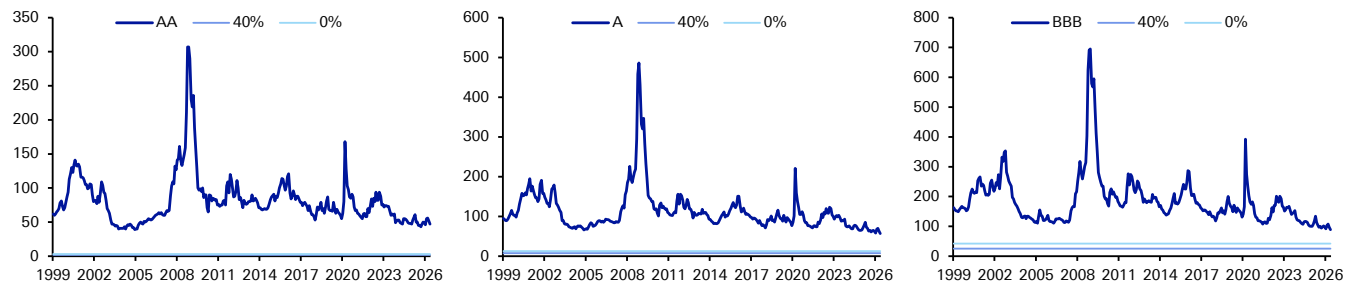
Investment Grade

Figure 63: EUR AA (left), A (middle) and BBB (right) Index Spread Histories vs. Default-Compensation Spreads (bps)



Source : Deutsche Bank, Markit Group

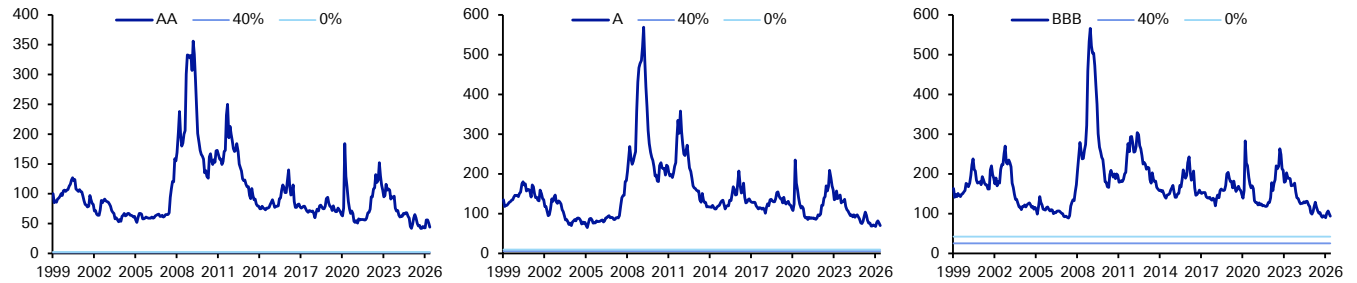
Figure 64: USD AA (left), A (middle) and BBB (right) Index Spread Histories vs. Default-Compensation Spreads (bps)



Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE Indices

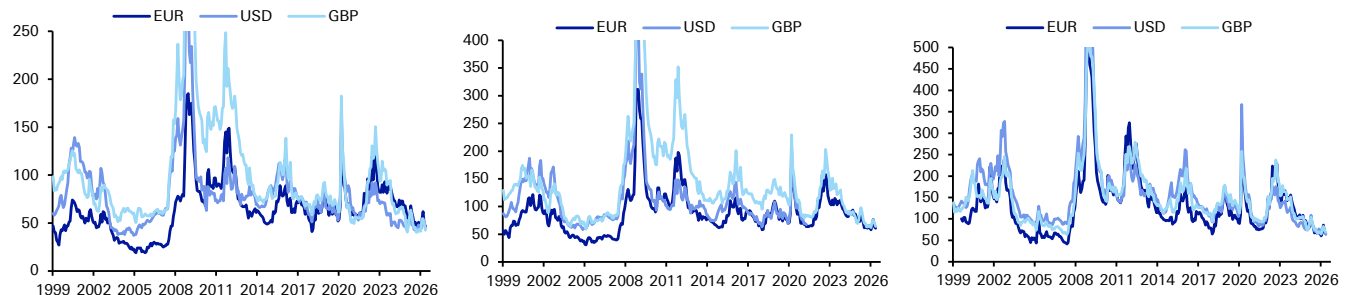


Figure 65: GBP AA (left), A (middle) and BBB (right) Index Spread Histories vs. Default-Compensation Spreads (bps)



Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE Indices

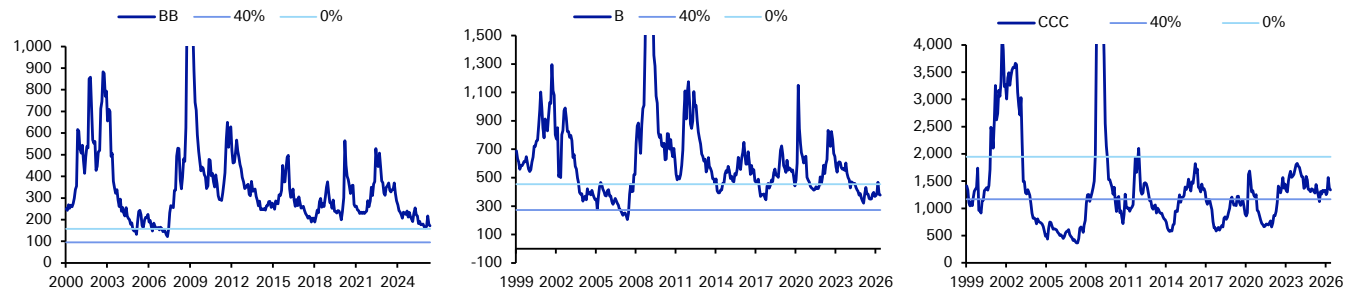
Figure 66: AA (left), A (middle) and BBB (right) Default Spread Premium (bps)



Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE Indices

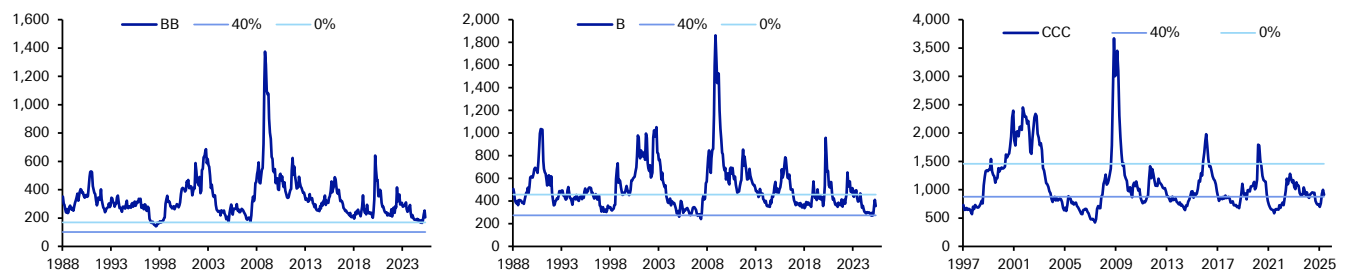
High Yield

Figure 67: EUR BB (left), B (middle) and CCC (right) Index Spread Histories vs. Default-Compensation Spreads (bps)



Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE Indices

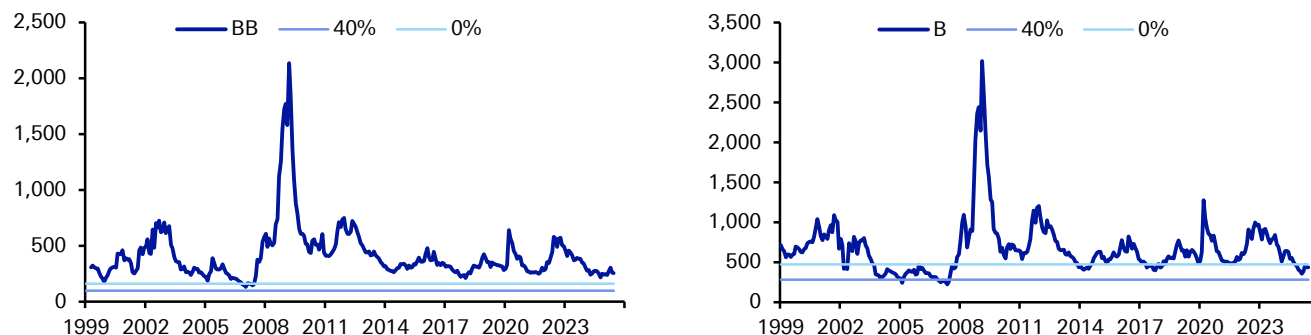
Figure 68: USD BB (left), B (middle) and CCC (right) Index Spread Histories vs. Default-Compensation Spreads (bps)



Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE Indices

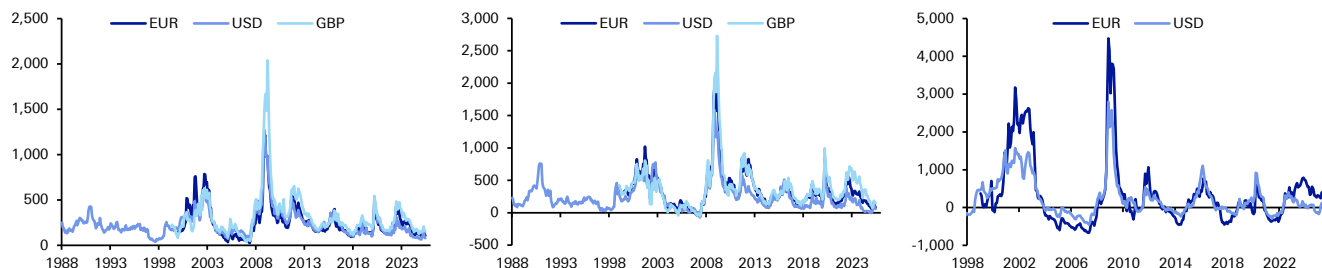


Figure 69: GBP BB (left) and B (right) Index Spread Histories vs. Default-Compensation Spreads (bps)



Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE Indices

Figure 70: BB (left), B (middle) and CCC (right) Default Spread Premium (bps)



Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE Indices

Why additional spread is required

This piece has only ever aimed to look at the spread required to compensate for default. In effect, it is the first building block in assessing where credit should trade, as defaults are the only way a buy-and-hold investor actually loses money in credit relative to the risk-free investment. In reality, there are other important factors that should be taken into consideration. However, trying to put a number to this becomes increasingly subjective and open to interpretation. For those wanting to explore further, these are some of the factors we believe should be considered for those not buying and holding to maturity.

Liquidity

While credit has always tended to be less liquid than government bonds (there were countries where this was not the case post the sovereign crisis), it could be argued this issue remains important largely due to the evolving regulatory environment and significant intervention in recent years. We have written on a number of previous occasions that liquidity risk is a potentially bigger worry for investors than default risk and why investors with a mark-to-market requirement may require more compensation. The initial sharp sell-off in credit during the pandemic arguably highlighted these liquidity issues. Although with the Fed buying corporate bonds for the first time after this shock, this might lead many to think the ultimate downside trading liquidity issue has been permanently reduced. However, what happens in the next crisis if inflation is higher?

Return over risk-free alternative



Clearly, one does not invest in credit to only get a return equal to the risk-free (i.e., government bonds). We need additional spread for it to be worth holding credit on a risk-reward basis.

What is the risk-free rate these days?

The sovereign crisis severely compromised the notion of a risk-free rate to benchmark credit. However, in the years since the sovereign crisis we've gone from high sovereign risk and high yields to one where more and more government markets were trading with negative yields as aggressive central bank intervention dominates. So, investors have had to question the suitability of government credit benchmarks from both extremes over the past few years. This makes it very difficult to assess relative risk. If inflation returns more permanently, this adds yet another dimension.

Cost of credit investing

Clearly, the further you move along the credit curve, the more intensive you need to be in analysing credit. Therefore, the cost base is higher and requires a higher reward than simply compensating you for default risk. Management fees also have to be paid out of returns.

Ratings transition risk

Some investors can only own bonds down to a certain rating category and may be forced to sell if a bond is downgraded through this lower limit. Hence, we probably need a premium for risk of downward rating migration, especially in troubled times. However, this changes over time and investors are less forced sellers on downgrades below certain levels than they used to be.

Uncertainty and timing of recovery rate

In an event of default, the amount and timing of recovery is uncertain. We note that ultimate recoveries for unsecured senior bonds have averaged 47.4% since 1987. This is above the level of 34.8%, over the same period, that trading prices a month after default implies. The latter is often the recovery rate used in models.

Short-term performance targets

Investment grade rarely fails to compensate for default risk over the medium to long run, but investors are very much measured on a short-term basis. They therefore need some cushion to protect against mark-to-market volatility.

Quantifying all these factors above is outside the remit of this report. The buy-and-hold investor has less need to worry about them, though. We hope this report shows the crucial default building block that credit spreads are built around.

Structural US & European corporate default models

Figure 71: US Structural Corporate Default Model (1983-2022, quarterly)

	Coefficients	t Stat	P-value
Intercept	-0.177	-7.1	0.0%
US Non Financial Corporate Debt to GDP (T-9)	0.386	7.7	0.0%
US Non Financial Corporate Short-Term/Long-Term Debt (T-21)	0.001	2.9	0.4%
BBB Real Yields (T-6)	0.005	5.3	0.0%
DB US Credit Recession Indicator (T-15, squared)	0	7.4	0.0%

Source : Deutsche Bank, Bloomberg Finance LP



Figure 72: Europe Structural Corporate Default Model (1983-2022, quarterly)

	Coefficients	t Stat	P-value
Intercept	-0.05	-2.2	2.9%
EA-20: Non Financial Corporate Debt to GDP (T-3)	0.001	2.9	0.5%
DB Eurozone Credit Recession Indicator (T-9, squared)	0	6.5	0.0%
Real ECB Policy Rate (T-3)	0.136	1.9	6.7%

Source : Deutsche Bank, Bloomberg Finance LP



Appendix 1

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